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A multi-objective mixed robust possibilistic flexible programming approach for sustainable seaport-dry port network design under an uncertain environment

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ABSTRACT

This paper develops a multi-objective mixed robust possibilistic flexible programming (MOMRPFP) approach for the sustainable dry port network design under uncertain environment. The optimal number, location, and capacity of dry ports, and the optimal number of containers transferred through dry ports are determined with minimizing the economic costs and environmental and social impacts. Finally, numerical analyses indicate that the total network cost decrease about 1.14% with the proposed approach. Moreover, the MOMRPFP approach improves the efficiency about the average computational time for large-sized networks with the presence of uncertainty in flexible objective functions and constraints as well as in data.

1. Introduction

The dry port concept is based on intermodal freight transport from inland terminals to seaports. From a logistics perspective, dry ports are designed to be an effective solution to increased traffic congestion, lack of storage space and the significant environmental impacts of seaports resulting from the growth in maritime container transport volume (Rodrigue, 2006; Roso and Lumsden, 2010). In addition, the dry port has been presented as an opportunity to improve seaport operations and the sustainability performance of international intermodal transport systems simultaneously (Hanaoka and Regmi, 2011; Roso, 2013). Because of these benefits, the dry port has garnered much attention from governments, port authorities, and researchers worldwide with the aim of achieving sustainable goals.

From a green logistics perspective, a green dry port is a sustainable hinterland port that not only meets the environmental requirements, but also raises economic interests (Hegazy, 2015; Iannone, 2012b). The concept of sustainable development strongly relies on three indispensable dimensions: economy, environment, and society. This concept has placed pressure on the construction and operation of dry ports today. Sustainable logistics networks, including groups of seaports and dry ports, are becoming increasingly crucial in sustainable intermodal freight transport chains (Bask et al., 2014; Vahdani et al., 2012). However, no relevant study has considered a sustainable dry port network (SDPN) that simultaneously pursues economic, environmental, and social objectives when operating dry port networks (DPNs) (Bentaleb et al., 2015).

SDPNs are becoming increasingly important in big countries (i.e., U.S.A., China, and Australia), where the exported cargo of hinterland origins is very far from seaports. The economic development policies related to the export of cargo have led a rise in transportation costs, CO_2 emissions, traffic jams, air pollution, and accidents (Hegazy, 2015). In that background, operation of DPNs is a solution to reduce the burden to the environment and society, and increase regional economic development. A study of SDPN

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design is necessary to contribute to the literature and practice by emphasizing the importance of economic, environmental, and social issues in DPNs. Accordingly, our research focuses on the optimal configuration of an DPN in consideration of sustainable development.

A SDPN related to environmental and social factors and the chaotic business environment (i.e., demand and cost fluctuation) impose a high degree of uncertainty in making decisions about the network design. Furthermore, most previous works have failed to consider uncertainty in designing SDPNs, which leads to higher risk in the long term. This paper evaluates more realistic situations in SDPN design by considering degrees of demand and cost uncertainties to bridge the aforementioned gaps. There are three main approaches for addressing uncertainty in the context of mathematical programming: stochastic programming, fuzzy programming, and robust programing (optimization). Robust optimization theory provides risk-averse methods for coping with uncertainty optimization problems. Our study uses a new robust optimization to address the multi-objective problem with business-as-usual uncertainty. We consider an export-oriented single product SDPN, consisting of a given single seaport, multiple potential locations for dry ports, and shippers, under uncertainties in demand, cost, capacity, carbon dioxide (CO_2) emissions, and social cost (i.e., unemployment and immigration costs, etc.). The objective is to determine the optimal number, locations, and capacity of dry ports, and the number of containers transferred through the dry ports, while minimizing the total network costs, including economic costs, CO_2 emission handling costs, and social costs.

The main contributions of this research from practical and theoretical perspective are as follows:

- Practical: This paper is the first one to consider DPN design problem in a sustainable and effective manner, where three indispensable dimensions of sustainability (economy, society, and environment) are considered under uncertain parameters in both objective functions and constraints. The proposed model and algorithm are expected to help the planners to determine the optimal number, location, and capacity of dry ports and the number of containers delivered to seaport through dry port by using the intermodal transports (road and rail transport). The CO_2 emissions are used for measuring environmental goal of transport modes and in opening dry ports of the proposed model, while the unemployment cost and immigration cost as well as the traffic congestion cost are considered for measuring social objective.
- Theoretical: This study proposes a multi-objective mixed robust possibilistic flexible programming (MOMRPFP) model, combining robust optimization and fuzzy programming (flexible programming and possibilistic programming) to formulate a SDPN design problem under uncertain environment. Subsequently, an interactive fuzzy multi-objective approach based on the modified ε-constraint method is proposed to solve the model. This approach would dominate other approaches used in previous studies owing to it can yield a high-quality solution with short computational time for a multi-objective optimization problem. The results indicate that the total network cost decreases about 1.14% and the average computational time is 5.33 s when compare to the α-parametric model for a fuzzy multi-objective programming problem. In addition, the MOMRPFP approach can also be used to design other large networks (i.e., supply chain networks) where the uncertain parameters are expressed as fuzzy numbers, with the efficiency of the average computational time of the proposed model (less than 2 min for test problem 2 with 6160 decision variables and 6396 constraints; less than 4 min for test problem 3 with 37,925 decision variables and 69,895 constraints).

The remainder of this paper is organized as follows. Section 2 presents a comprehensive literature review of DPNs, sustainable development, and distribution network design. The research problem is defined and formulated in Section 3. In Section 4, we propose an MOMRPFP approach for solving the problem. Numerical analysis is presented in Section 5. Concluding remarks and suggestions for future research are provided in Section 6.

2. Literature review

This section begins with a summary of the concept and role of dry port. In the second part, sustainability in distribution network design is discussed. Finally, fuzzy approaches to overcoming uncertainty in distribution network designs are outlined.

2.1. Dry port concept

Freight transport through the use of standard containers has witnessed a continuous increase in the capacity and volume of vessels since the 1960s. This has led to bottlenecks in the inland transport system and a lack of space at port terminals in most seaports (Roso and Lumsden, 2009). Dry ports have been considered an effective solution to this problem since the 1980s because of their supporting for the container traffic of seaports and production-distribution chains (UNCTAD, 1982). The dry port concept was first mentioned in the 1982 United Nations report, where a terminal to (or from) which shipping companies could issue their own bills of lading was described as an inland port for all import (or export) cargo. Cullinane et al. (2012) and Roso and Lumsden (2009) have also defined a dry port as an inland port directly connected to one or several seaports with high capacity transport means.

Based on the intermodal freight transport systems, dry ports are nodes in transport network that separate the various links of the transport chain to obtain the shortest possible routes to be used by the lowest capacity and most polluting transport means (Hanaoka and Regmi, 2012; Roso, 2007). This leads to a greater efficiency in operations and energy use of transport chains. In addition, dry ports directly connect to seaports with high capacity transport modes, preferably rail, to prevent traffic bottlenecks in hinterland transport and reduce and lower environmental impacts, such as those of road and railway emissions. A case study in Lättilä et al.

(2013) demonstrates that CO_2 emissions and external costs can be reduced to 28% and 35%, respectively, when the dry port is implemented in transport. Dry ports are also an effective solution for reducing the related transportation costs by increasing the number of dry ports in the intermodal transport network (Henttu and Hilmola, 2011). Crainic et al. (2015) provided a dry port concept based on the optimal design and management of a freight distribution network, according to which a dry port was defined as a hinterland distribution center in a freight distribution network with intermodal transport systems.

The dry port concept presents opportunities to support seaport operations by solving the problems of seaports (i.e., congestion, lack of storage space, high CO2 emissions in the seaports), which is caused by the uncontrolled increase of road transportation of containers. With DPNs, it is possible to reduce the number of accidents, level of congestion, and noise in the seaports by taking advantage of the economies of scale of high capacity transport modes, especially rail (Lättilä et al., 2013). Dry ports enable seaports to reach new potential customers by increasing their inland access and reducing traffic congestion in and around seaports because the number of road links to and from the seaports is reduced (Feng et al., 2013). To enhance the competitiveness of seaports, Bask et al. (2014), Beresford et al. (2012), Andersson and Roso (2016), Iannone (2013), and Roso and Rosa (2012) defined a dry port as an extended gateway concept which offers a diverse set of value-added logistics services for shippers (i.e., labeling, warehousing, customer clearance, packing, and cross docking) that are usually available at seaports.

The green or sustainable dry port concept has appeared in recent works because of its significant contributions to environmental and social issues. Bergqvist and Zandén (2012) and Iannone (2012b) have defined a green dry port as an efficient, and environmentally and socially friendly hinterland port supporting the container traffic of seaports, whereas a sustainable dry port emphasizes efficiency and economy, as well as concern for both environmental and social factors. Both green and sustainable dry ports aim to reduce harmful effects on the environment and society, but the sustainable dry port is a broader concept where intangible social factors are considered for sustainable goals in the future. A SDPN is a network that includes several dry ports directly connected to one or more seaports by high capacity transportation with an aim to sustain development by minimizing potential harmful effects on the environment and society while also ensuring economic benefits.

Dry ports are classified into three types based on their function and location relative to the seaport: distant dry ports, midrange dry ports, and close dry ports (Roso et al., 2009). Fig. 1 shows the structure of a DPN with three types of dry ports. This figure suggests that a close dry port offers greater possibilities for buffering containers around a seaport to reduce local traffic problems and the lack of storage space. The midrange dry port serves as a consolidation point for different rail flows from shippers, and supports administration and technical equipment specific for seaports. In this study, the distant dry ports are considered as supporting regional economic development and evaluating the effects of long distance transport on the environment and society.

2.2. Sustainability in distribution network design

Sustainable development concept plays a fundamental role in current distribution networks as the potential harmful effects of hinterland transport become increasingly understood (Bask et al., 2014; Crainic et al., 2015). The influence of networks must be determined in relation to the three main dimensions of sustainability: economic performance, environmental impact, and social responsibility.

Economic performance is one of the most crucial factors, and is the ability to concurrently create sustainable profit (or stable operating cost) in the future and maintain a balance between the public goals of the government and private benefits of port authorities when establishing networks. In the port-hinterland network design literature, there are many studies that have considered single objectives to evaluate the economic performance of the system, such as minimizing total costs (Crainic et al., 2015; Chang et al., 2015; Feng et al., 2013, Joong et al., 2018) or maximizing total profits (Qiu and Lam, 2014; Qiu et al., 2015).

Environmental impacts are adverse effects caused by industrial development, infrastructural projects, or the release of substances into the environment. Road transport not only plays a critical role in the development of freight export networks but is also responsible for by far the largest share of environmental problems. The CO_2 emissions volume, pollution, and noise are factors that have

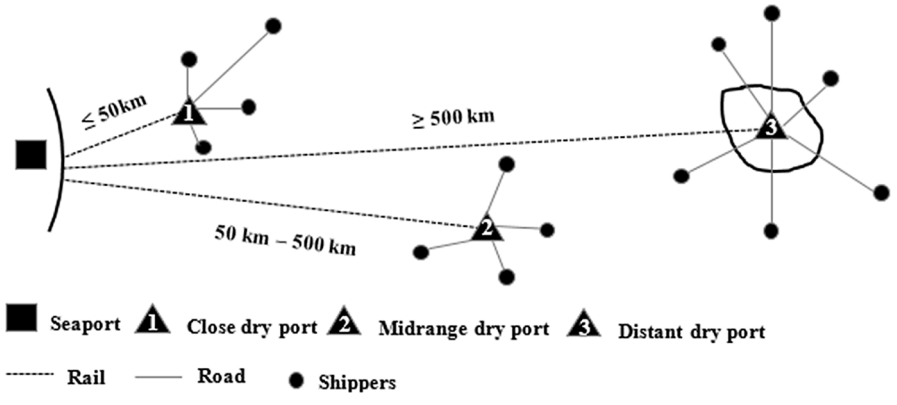


Fig. 1. Network flows on a dry port network with three types of dry port.

been considered in some dual-objective programming models of economy and environment to reduce negative environmental effects through port hinterland network design (Ambrosion et al., 2016; Gu and Lam, 2013; Janic, 2007).

Despite the importance of social issues in port-hinterland networks, the relevant literature is not wide. As a primary work in this area, Iannone (2011) and (2012a) made a significant step forward in incorporating the concept of "social cost" in port-hinterland networks. Lättilä et al. (2013) used tangible indicators, referred to as "external costs" in transport systems, such as noise, congestion, and pollution costs, to evaluate both the environmental and social aspects of networks. A systematic review by Bentaleb et al. (2015) of dry port development between 1986 and 2015 revealed that there is a paucity of studies involving sustainability in the field of mathematical programming.

To the best of our knowledge, no research has comprehensively considered the three dimensions of sustainability in DPN design. To address the above-mentioned gap in the SDPN design literature, this research tackles the socially responsible DPN design problem where three objective including economy, environment and society are considered in multi-objective programming problem.

2.3. Uncertain programming approaches in distribution network design

Despite the importance of the uncertainties in the design and operation of distribution networks, especially when social factors are involved, the relevant literature is not wide in the DPN design field. In this section, we review the solution approaches, such as stochastics programing, fuzzy programing, and roust programing in the distribution network design field, especially supply chain network design.

Mousazadeh et al. (2014) classified the uncertainty into two groups: (1) the uncertainty in data may be caused by the random nature of the parameters and (2) the lack of information regarding the exact values of the parameters. Stochastic programming approaches are mostly applied in the area as An and Lo (2016), Yang et al. (2016) and Garrido et al. (2015), when the probability distribution of random variables is predetermined. Nevertheless, in many real cases, the distributional information of random data is unknown because of the lack of historical data, and robust programming methods are combined to address this problem. Some eminent studies in this area are those of An and Lo (2015), Ghodratnama et al. (2015) and Shabani and Sowlati (2016).

Fuzzy programming provides a framework to handle different kind of uncertainty, which the uncertain parameters are expressed as a suitable possibility distribution in the fuzzy number form based on the opinions of decision makers. Fuzzy approaches reduce time and resource to determine the probability distribution of uncertain parameters as well as a large number of scenarios used in representing the uncertainty in stochastic approaches. Many interactive fuzzy multi-objective approaches have been developed to solve the location-allocation problem in logistic network design and supply chain management-related problems (Mousavi et al., 2014; Niakan and Rahimi, 2015; Pishvaee and Razmi, 2012; Subulan et al., 2015; Vahdani et al., 2013, Tsao et al., 2018).

Robust programming provides powerful risk-averse methods to deal with epistemic uncertainty in optimization problems. The extension of robust optimization into fuzzy programming is proposed to fully utilize the advantages of robust programming in the fuzzy mathematical programming area. To the best of our knowledge, no recent research applies robust optimization into a fuzzy programing approach in the context of SDPN design. However, a few studies use the robust optimization approach for designing supply chain networks. Pishvaee et al. (2012) first applied robust possibilistic programming based on the expected intervals and values of trapezoidal fuzzy numbers to solve the location-allocation problem for socially responsible supply chain networks. Zahiri et al. (2014) proposed robust possibilistic programming to solve a multi-period location-allocation problem, where the uncertain parameters are handled by fuzzy programming. Bairamzadeh et al. (2015) was the first to develop a robust possibilistic programming approach for a multi-objective problem having uncertain parameters in objective functions, and applied it to sustainable bioethanol supply chain design. A novel robust stochastic-possibilistic programming was proposed in Ehsan et al. (2018) to cope with hybrid uncertain parameters when designing closed-loop supply chain network.

In the context of DPN design, some genetic algorithms are applied to find near optimal solution for the locations selection of dry ports (Chang et al., 2015). Also, there are other works using fuzzy multi-criteria decision-making models such as Fuzzy C-Means Clustering to select the locations for dry ports in the network (Li et al., 2011; Li and Jiang, 2014; Núñez et al., 2014; Wei et al., 2010). No research has applied robust programming approaches for making strategic and tactical decisions when designing DPNs under uncertain environment. To address the above-mentioned gap in the SDPN literature, this research develops a mixed robust possibilistic flexible programming model for a multi-objective SDPN design problem at the same time under uncertain parameters in both objective functions and constraints.

3. Problem description and formulation

In this section, we first present an original SDPN model for representing the considered problems in this study and then describe a mathematical model formulation for the design of an SDPN.

3.1. Problem definition

An export-oriented SDPN for single freight is considered to comprise a single seaport, multiple potential locations for dry ports, and shippers. Links connect the shippers to the seaport, either directly by road or through dry ports by rail. Fig. 2 shows the structure of the SDPN. The government makes strategic decisions about the number, locations, and capacity of dry ports, while shippers make

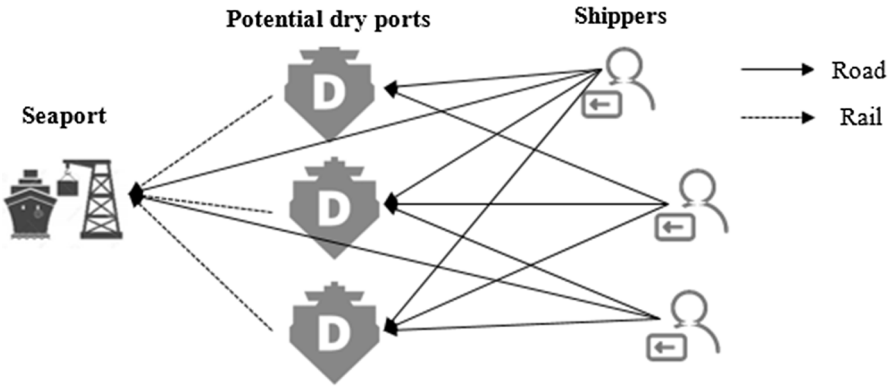


Fig. 2. Network flows on a dry port network.

mid-term decisions about the route and number of freight containers traveling to the seaport, either directly or through the dry ports. Therefore, our model is an integrated strategic-tactical one. Freight allocation is considered on the basis of the established capacity for each dry ports. In addition, the following assumptions are made:

- All trucks are identical and all containers are homogeneous. It means that we consider one type of 400 foot container (2 TEU) for single exported freight and one truck carries one container.
- The location of the single seaport, multiple candidate sites for installing dry ports, and shippers are predetermined.
- The opening cost and the number of created job opportunities are dependent on various given capacity levels of each dry port.
- The freight volume of each shipper is known and must be exported through the seaport.
- The established dry ports contribute to immigration problems in the area.
- The CO_2 emissions depend on installed capacity at each dry port, transportation model, fuel usage, and geographical distances.

In our model, the SDPN is considered from the whole system perspective for sustainable goals (economy, environment, and society). From the sustainable development perspective, all different stakeholders (i.e., government, seaport, dry port operators, and shippers) participating in a SDPN are responsible for minimizing the total cost of the whole network including economic, environmental, and social costs (Hegazy, 2015). Therefore, minimizing the total costs of the DPN including economic, environmental, and social cost are chosen as the problem's objective.

Economic performance is related to the efficiency of the SDPN design in operational costs or future profits. Total cost (i.e., set up cost and transportation, storage, and holding cost) is usually minimized in evaluating the economic efficiency of logistic and distribution networks. In the context of increasing private sector participation in reducing the financial burden on the government, economic costs must be considered in this study to ensure a suitable balance in the public-private partnership between the government and the dry port operators regarding social and environmental problems and profit (Rodrigue and Notteboom, 2009; Rodrigue et al., 2010).

Logistic activities are recognized to have an environmental contribution to climate change namely greenhouse gas emissions (Bergqvist and Zandén, 2012; Roso and Lumsden, 2009; Roso et al., 2009). Among these, transportation is responsible for one-fourth of the CO_2 emission volume and is still growing. The environmental effects of transportation, particularly CO_2 emissions, can be reduced by replacing transport flows from road to intermodal solutions. Dry port networks that directly connect to seaports and shippers by rail are considered in this study to be an efficient solution for reducing the negative effects of CO_2 emissions on the sustainability of DPNs and the environment. CO_2 emissions are a well-known environmental measure that has previously been used in the transport sector to evaluate the environmental burden of the network. In our model, we inspect the CO_2 emissions handling costs in the establishment and operation of dry ports as well as in all transport flows when designing the SDPN.

Social costs are closely associated with traffic congestion and accidents and human health hazards derived from the increasing growth of road container transport in developing countries (Bask et al., 2014; Roso, 2013). Road transport plays a crucial role in hinterland economic development, and the combination of road and rail transport in DPNs is an effective solution for reducing harmful effects on the society. Several standards, such as ISO 26000 (2010) and SA8000 (2001), have been developed to support the planning and implementation of corporate social responsibility (CSR) in logistics service providers, referred to as logistic social responsibility (Carter and Jennings, 2002). In the DPN design context, Iannone (2012b) examined the social costs of port-hinterland networks associated with congestion, climate change, pollution, noise, and accidents that derive from transport operations. A theoretical framework for the growing attention to both hinterland logistics and CSR was provided by Bergqvist and Zandén (2012), who investigated the affected stakeholders, i.e., the port, society, transport service providers, and shippers, to implement CSR in port-hinterland networks. Based on the public-private partnership perspective of Rodrigue et al. (2010) and Chang et al. (2018), we model the social costs which include two components: (1) government investments (public perspective) for regional development ensuring

quality of life and the safety of society, such as health and insurance costs; and (2) the damage of shippers (private perspective) caused by harmful effects of establishing DPNs, such as traffic congestion and accident costs. These social costs affect the location-allocation decisions in a SDPN. Therefore, all stakeholders should consider them while making their decisions to ensure sustainability.

Accordingly, three relevant social measures are selected as social measures in the proposed model including: (1) the cost of unemployment, (2) the cost of immigration, and (3) the cost of traffic congestion time of transport flows. The first measure represents the community development from the creation of more job opportunities to minimize the social costs of unemployment. The importance of this measure has been highlighted previously in the supply chain management context (Pishvaee et al., 2012). The second measure reflects the social risks that could develop from the immigration problem caused by opening the dry ports. This measure has not been mentioned in the previous literature on CSR in the logistics sector. However, it is known as a credible measure for ensuring quality of life and the safety of society by the community involvement and development group of ISO 26000. Notably, in the current problem, the decision about capacity highly influences the number of created jobs and the impact of immigration issues. The third measure is a well-known social measure that has previously been applied to measure the social costs of a logistics network, especially for transport activities.

A multi-objective mathematical model for optimizing SDPN under an uncertain environment of cargo export demand, CO_2 emissions caused by transport activities, and social costs related to traffic congestion, unemployment, and immigration is proposed in our works.

3.2. Model formulation

The indices, parameters and variables used to formulate the concerned SDPN design problem are illustrated as follows.

Indices

P	set of shippers
D	set of candidate locations of dry port
M	set of capacity levels for candidate dry ports

Parameters

cf_d^m	fixed cost of opening a dry port d with capacity level m [USD]
c^{rd}	transportation cost per container per distance by road [USD per container per kilometer]
c^{rl}	transportation cost per container per distance by rail [USD per container per kilometer]
c_d^{str}	cargo storage cost at dry port d with capacity level m per day [USD per container per day]
c_d^{handl}	cargo handling cost at dry port d with capacity level m [USD per container]
c^{hc}	cargo holding costs in transport process [USD per container per hour]
c^{hr}	time unit cost of resources including human and vehicle [USD per hour]
f^{id}	distance transport from shipper i to dry port p [kilometer]
f^{ds}	distance transport from dry port p to sea port s [kilometer]
f^{is}	distance transport from shipper i to sea port s [kilometer]
f^{im}	parameter that depends on immigrant rate of each area [percentage]
n^{uep}	number of the unemployment before candidate dry port d is opened [number]
n^{ep}	number of the created employment when candidate dry port d is built [number]
o^{op}	carbon dioxide emissions for opening and operating a dry port d with capacity level m [kg eq per container]
o^{rd}	carbon dioxide emissions per container per distance by road [kg eq per container per kilometre]
o^{rl}	carbon dioxide emissions per container per distance by rail [kg eq per container per kilometre]
e^{cb}	unit cost of carbon dioxide emissions handling in the outside market [USD per kg eq]
q_{ds}^m	volume of containers transported from candidate dry port d with capacity m to seaport s [Container]
s^{uep}	social cost for the per unemployment [USD per person]
s^{im}	social cost for per immigrant [USD per person]
r_i	exported cargo demand of shipper $i \in P$ [Container]
T_d	operational time of candidate dry port d [day]
t_d	cargo storage time at dry port d with capacity level m [day]
t^{dl}	average traffic congestion time of each area [hour per kilometer]
φ_d^m	capacity with level m for candidate location of dry port d [Container]
ψ_{eco}	weighting factor of the cost components in the economic goal [Percentage]
ψ_{env}	weighting factor of the cost components in the environmental goal [Percentage]
ψ_{soc}	weighting factor of the cost components in the social goal [Percentage]

Decision variables:

$x_d^m =$	$\begin{cases} 1 & \text{if dry port } d \text{ with capacity level } m \text{ is opened} \\ 0 & \text{otherwise} \end{cases}$
q_{id}^m	volume of containers transported from shippers $i \in P$ to candidate dry ports $d \in D$ with capacity $m \in M$
q_{is}	volume of containers transported directly from shippers $i \in P$ to seaport $s \in S$

$$\text{Min } W_1 = \psi_{eco}\left(\sum_{d\in D}\sum_{m\in M}\tilde{c}f_d^m x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\tilde{c}^{rd}f^{id}q_{id}^m + \sum_{d\in D}\sum_{m\in M}\sum_{i\in P}(\tilde{c}_d^{str}t_d + \tilde{c}_d^{handl})q_{id}^m + \sum_{d\in D}\sum_{m\in M}\tilde{c}^{rl}f^{ds}q_{ds}^m\right) + (1 - \psi_{eco})\sum_{i\in P}\tilde{c}^{rd}f^{is}q_{is} \tag{1}$$

$$\text{Min } W_2 = \psi_{env}\left(\sum_{d\in D}\sum_{m\in M}\tilde{o}^{op}x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\tilde{o}^{rd}f^{id}q_{id}^m + \sum_{d\in D}\sum_{m\in M}\tilde{o}^{rl}f^{ds}q_{ds}^m)\tilde{e}^{cb}\right) + (1 - \psi_{env})\sum_{i\in P}\tilde{o}^{rd}f^{is}q_{is}\tilde{e}^{cb} \tag{2}$$

$$\text{Min } W_3 = \psi_{soc}\left(\sum_{d\in D}\sum_{m\in M}(n^{uep} - \tilde{n}^{ep})s^{uep}x_d^m + \sum_{d\in D}\sum_{m\in M}s^{im}\tilde{f}^{im}\tilde{n}^{ep}x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}(\tilde{c}^{hr} + \tilde{c}^{hc}q_{id}^m)\tilde{t}^{dl}\right) + (1 - \psi_{soc})\sum_{i\in P}(\tilde{c}^{hr} + \tilde{c}^{hc}q_{is})\tilde{t}^{dl} \tag{3}$$

In this model, the overall economic costs, CO_2 emissions handling cost, and social costs of dry ports and shippers are considered through three sustainable goals. The weighting factors ψ_{eco}, ψ_{env}, and ψ_{soc} in range [0, 1] indicate the relative importance between the dry port's costs and shipper's costs in each economic, environmental, and social objective function, respectively.

For economic consideration, the first objective function (1) minimizes the total economic cost, which consists of setup cost, storage and handle cost at dry ports, and transport cost between hinterland origins and the seaport through dry ports. The setup cost depends on the established capacity levels at candidate dry ports. The storage costs include preserving and storing cost, depending on the number of containers (q_{id}^m) and the cargo storage time (t_d) at dry ports. The handling cost includes the loading and unloading costs, which depends on the number of containers (q_{id}^m) at dry ports. Finally, transport cost occurs based on the number of transported containers and distance between facilities (shippers, dry ports, and sea port).

The second objective function (2) minimizes the environmental impacts determined by the CO_2 emission index related to opening and operating of dry ports as well as transport activities. The CO_2 emission caused by opening and operating of dry ports is presented in the first term. It is estimated based on the capacity established at each dry port. The three last terms express the CO_2 emission related to road and rail transport from shippers to dry ports and sea port. The unit CO_2 emission per distance travelled by road or rail transport is estimated according to vehicle speed and fuel consumption (Xu et al., 2013). In practice, some assumptions are used for this estimation. For example, in our test problems, we assume that diesel fuel is used for both transport modes, and the vehicle speed is derived from their design parameters. The uncertain parameter regarding CO_2 emission handling cost is used to evaluate total environmental cost in the model.

Social responsibility is addressed in the third objective function (3), wherein the first and second terms relate to the unemployment and immigration cost per laborer. When the dry ports are established, they not only contribute to new job opportunities for local people but also attract immigration from other areas. If the number of job opportunities created is maximal, the unemployment cost is reduced, but the immigration cost (e.g., management cost, insurance cost) may rise because of its attractiveness to immigrants. The decision about the location and capacity of dry ports should consider a reasonable trade-off between unemployment cost and immigration cost to obtain an equal balance in the SDPN. The other terms are traffic congestion cost caused by freight transport from the shipper to the dry port and seaport. The traffic congestion cost in the model comprises two components. The first is the fixed time cost of resources human and vehicle and the second is cargo holding cost, which depends on the volume of containers transported and average traffic congestion time (t^{dl}) in each area of the network. t^{dl} is related to the location of candidate dry ports. In general, t^{dl} is determined based on the basis of roughly estimated queue lengths and mean queue speed by means of observations of highway policemen (Hansen, 2001).

$$t_d = \frac{T_d}{\tilde{\varphi}_d^m} \tag{4}$$

$$\sum_{d\in D}\sum_{m\in M}q_{id} + q_{is} = r_{i\in P},\ \forall\, i \in P \tag{5}$$

$$\sum_{i\in P}q_{id} = q_{ds},\ \forall\, d \in D \tag{6}$$

$$\sum_{i\in P}q_{id} \leqslant \tilde{\varphi}_d^m x_d^m,\ \forall\, d \in D, \forall\, m \in M \tag{7}$$

$$\sum_{m\in M}x_d^m \leqslant 1,\ \forall\, d \in D \tag{8}$$

$$x_d^m \in \{0; 1\},\ \forall\, d \in D, \forall\, m \in M \tag{9}$$

$$q_{id} \geqslant 0, \forall\, i \in P,\ \forall\, d \in D \tag{10}$$

$$q_{is} \geqslant 0, \forall\, i \in P, \quad \forall\, s \in S \tag{11}$$

Eq. (4) is used to estimate the cargo storage time at a dry port. It is related to the decision on capacity levels established (φ_d^m) at candidate dry ports. When the installed capacity is great enough, the cargo storage time will decrease. Constraint (5) ensures that the cargo export demand of each hinterland origin is satisfied through two ways to route freight to seaport, either directly or through dry ports. Constraint (6) strikes a balance between the input and output of each dry port. Constraint (7) is a constraint on the capacity of dry ports. The right-hand-side of this constraint ($\varphi_d^m x_d^m$) prohibits the transport of cargo unit between shippers and dry ports that have not yet opened. Constraint (8) ensures that no more than one capacity level be assigned to each dry port in a given location. Finally, constraints (9), (10), and (11) guarantee the binary and non-negativity of decision variables.

The binary variable x_d^m and the continuous variables q_{id}^m and q_{is} in the model are designed to solve the research questions about the number, locations and capacity selection of dry ports as well as the optimal volume of containers through dry ports, respectively. The symbol "~" is used to assign uncertain parameters in business-as-usual related to cargo export demand and some relevant economic, environmental, and social costs in the objective functions and system constraints, which is modeled using triangular fuzzy numbers.

4. The proposed solution method

In this section, we propose a MOMRPFP model for the SDPN design problem based on the extension of mixed possibilistic flexible programming model proposed by Pishvaee and Khalaf (2016). However, their model is used to solve a single objective optimization problem. To develop the MOMRPFP model for a multi-objective optimization problem, we consider three concepts of a robust solution (i.e. average performance, feasibility robustness, and optimality robustness) as separate objective functions in the proposed model to control the satisfaction degree of each objective function and the violation of each flexible constraint in the final preferred compromise solution. A solution to an optimization problem is defined as its robustness if it has both optimality and feasibility robustness. Optimality robustness means that the value of objective function remains near optimal for all scenarios of uncertain input data. A solution's feasibility robustness means it remains feasible for all possible values of an uncertain parameter (Pishvaee et al., 2012). Our approach differentiates our contribution from previous studies because the penalty cost function is considered as an objective functions (OF_3) in our model to control the violation of soft constraints. We assume that the capacity constraint of facilities is soft due to the constraints are hard to hold for all feasible solutions. Because the soft constraints are associated with penalty cost functions which are used to develop the MOMRPFP model. In multi-objective optimization problems, the penalty functions of soft constraints will be a balance or compromise between competing goals. In addition, the use of soft constraint will significantly reduce a large number of uncertain scenarios that other popular uncertain programming methods often encounter, such as scenario-based stochastics programming.

Also, in many real cases, the parameters such as demand and cots are affected by many factors. For example, the traffic congestion cost depends on the volume of containers transported and average traffic congestion time. In practice, these parameters could be accurately measured. However, to achieve accurate values of these parameters consume a large amount of time and resources. Also, the measurements may be inaccurate or particular data may be deliberately hidden. Thus, we consider that the parameters have epistemic uncertainty in the model to cope with uncertain conditions (Mousazadeh et al., 2014; Pishvaee et al., 2012). Our MOMRPFP approach takes advantage of fuzzy possibilistic flexible programming to effectively cope with epistemic uncertainties as well as flexibility in constraints and goals simultaneously (Dubois et al., 2003).

Robust fuzzy programming can be categorized into (1) robust flexible programming and (2) robust possibilistic programming based on two main branches of fuzzy mathematical programming (flexible programming and possibilistic programming). A robust flexible programming model is used to cope with the uncertainty in flexible objective functions and constraints while robust possibilistic programming model is applied to deal with the uncertainty in data (Pishvaee and Khalaf, 2016). The MOMRPFP approach is an extension of robust programing in two branches of fuzzy programming to completely consider uncertain resources for optimizing

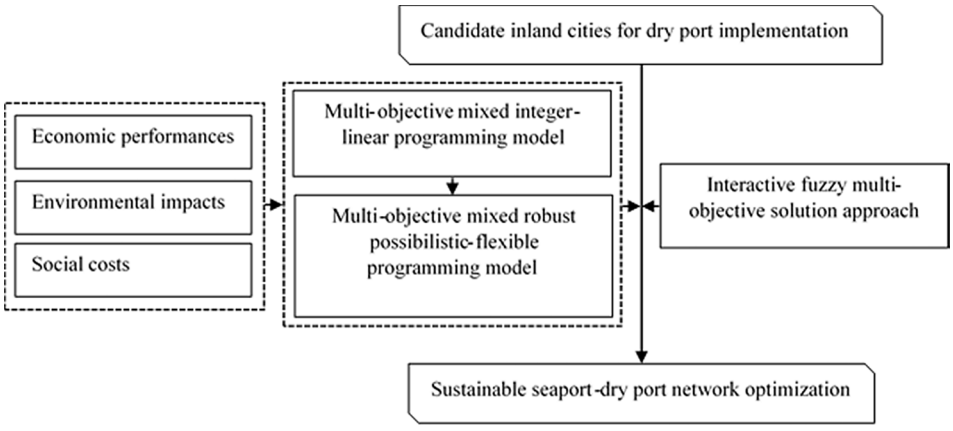


Fig. 3. Decision-making framework for sustainable dry port network design.

multi-objective programming problems. Fig. 3 depicts the SDPN design decision-making framework employed in this research. To solve this MOMRPFP model, we have proposed a fuzzy solution method based on the ε-constraint method to find the final preferred compromise solution.

4.1. Mixed possibilistic flexible robust programming model

The compact form of the mixed integer–linear programming model for the economic objective function (excluding the environmental and social objective functions) of an SDPN can be represented as follows.

$$\begin{aligned}
&\text{Min } W_1 = \widetilde{f}\, x_d^m + \widetilde{c}\,(q_{id} + q_{is}).\\
&\text{S. t}\\
&t_d = \tfrac{T_d}{\widetilde{\varphi}_d^m},\\
&Aq_{id} + Bq_{is} = \widetilde{r}_i,\\
&Cq_{id} = 0,\\
&Sq_{id} \leqslant \widetilde{\varphi}_d^m x_d^m,\\
&Px_d^m \leqslant 1,\\
&x_d^m \in \{0, 1\},\\
&q_{id},\, q_{is} \geqslant 0.
\end{aligned} \tag{12}$$

According to model (12), vectors f and c are related to fixed costs of opening dry ports and variable activity costs. The matrices T, A, B, C, S, and P represent the constraints' coefficients, φ_d^m is the soft constraints' coefficients regarding capacity of dry ports. Vectors r_i and t_d are representative of the shippers' freight export demand and the storage time of freight in each dry port, respectively. Finally, vector x is a binary variable and q_{id} and q_{is} represent continuous variables. According to the Pishvaee and Khalaf (2016) the mixed possibilistic flexible robust programming (MPFRP) model for single objective can be formulated as follows:

$$\begin{aligned}
&\text{Min } W_1 = \left(\tfrac{f^{(1)}+f^{(2)}+f^{(3)}}{3}\right)x_d^m + \left(\tfrac{c^{(1)}+c^{(2)}+c^{(3)}}{3}\right)(q_{id} + q_{is})\\
&+ \pi\left[\left(f^{(3)}x_d^m + c^{(3)}(q_{id} + q_{is})\right) - \left[\left(\tfrac{f^{(1)}+f^{(2)}+f^{(3)}}{3}\right)x_d^m + \left(\tfrac{c^{(1)}+c^{(2)}+c^{(3)}}{3}\right)(q_{id} + q_{is})\right]\right]\\
&+ \omega\left[r^{(2)} - \delta\left(\tfrac{r^{(2)}+r^{(3)}}{2}\right) + (1-\delta)\left(\tfrac{r^{(1)}+r^{(2)}}{2}\right)\right] + \theta\left[\rho\left(\tfrac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}\right) + (1-\rho)\left(\tfrac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}\right) - \varphi_d^{m(1)}\right]x_d^m\\
&+ \chi\left[\left(\tfrac{z^{(1)}+z^{(2)}+z^{(3)}}{3}\right)(1-\alpha)\right] + \eta\left[\left(\tfrac{v^{(1)}+v^{(2)}+v^{(3)}}{3}\right)(1-\beta)\right]x_d^m.\\
&\text{S. t}\\
&t_d = \frac{T_d}{\rho\left(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}\right) + (1-\rho)\left(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}\right) + \left(\frac{v^{(1)}+v^{(2)}+v^{(3)}}{3}\right)(1-\beta)},\\
&Aq_{id} + Bq_{is} = \left[\delta\left(\tfrac{r^{(2)}+r^{(3)}}{2}\right) + (1-\delta)\left(\tfrac{r^{(1)}+r^{(2)}}{2}\right)\right] - \left(\tfrac{z^{(1)}+z^{(2)}+z^{(3)}}{3}\right)(1-\alpha),\\
&Cq_{id} = 0,\\
&Sq_{id} \leqslant \left[\rho\left(\tfrac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}\right) + (1-\rho)\left(\tfrac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}\right)\right]x_d^m + \left[\left(\tfrac{v^{(1)}+v^{(2)}+v^{(3)}}{3}\right)(1-\beta)\right]x_d^m,\\
&Px_d^m \leqslant 1,\\
&x_d^m \in \{0, 1\},\\
&q_{id},\, q_{is} \geqslant 0.
\end{aligned} \tag{13}$$

In the model (13), the first and second terms of the objective function denote the average economic performance of the concerned system based on the expected value of uncertain parameters. The third term minimizes the maximum deviation over the expected value of the objective function by adjusting the degree of optimality robustness. The parameter π determines the significance of this term against the other terms. All remaining terms control the feasibility robustness by minimizing total penalty costs stemming from possible violation of soft constraints. The parameters ω, χ, η, and θ are representative of the penalty cost for each unit of violation of constraints resulting from the imprecision of uncertain parameters. Two triangular fuzzy numbers (z and v) can be used to show the violation of soft constraints which are related to the freight export demand and capacity of the dry ports, respectively. The parameters δ and ρ denote the confidence level of constraints delimiting uncertain parameters ($0.5 < \delta, \rho \leq 1$), and the parameters α and β are defined as the level to which soft constraints are satisfied. The values of the aforementioned parameters should be assigned by decision makers.

The MPFRP model is able to efficiently determine the optimal values of confidence levels as well as to control the deviation of objective function values from the expected values of each objective function. This process is also used to transform the second and the third objective functions (W_2 and W_3) in the original model into the MRFRP model.

4.2. Multi-objective mixed robust possibilistic flexible programming model

The base model includes three objective functions related to the economy, environment, and society of the SDPN, which can be

stated as follows:

$$\begin{aligned}
&\text{Min } W_1 = \widetilde{f}\, x_d^m + \widetilde{c}\,(q_{id} + q_{is}) \\
&\text{Min } W_2 = \widetilde{o}\,(q_{id} + q_{is}) \\
&\text{Min } W_3 = \widetilde{s} x_d^m + \widetilde{k}\,(q_{id} + q_{is}) \\
&\text{S. t} \\
&t_d = \tfrac{T_d}{\widetilde{\varphi}_d^m}, \\
&Aq_{id} + Bq_{is} = \widetilde{r}_i, \\
&Cq_{id} = 0, \\
&Sq_{id} \leqslant \widetilde{\varphi}_d^m x_d^m, \\
&Px_d^m \leqslant 1, \\
&x_d^m \in \{0, 1\}, \\
&q_{id}, q_{is} \geqslant 0.
\end{aligned} \tag{14}$$

Here, vectors *o, s,* and *k* are uncertain parameters that correspond to unit carbon emission, social cost (i.e., unemployment and immigration cost), and congestion cost resulting from the establishment of the dry ports and the transport activities through the DPN.

As expressed in the previous section, the robust objective function must consider the three components: maximizing the expected performance of the concerned network, minimizing optimality robustness, and maximizing feasibility robustness. Therefore, to develop the MOMRPFP model, the aforementioned concepts are considered as separate objective functions in the proposed model.

To maximize the expected sustainable performance of the SDPN, the Torabi and Hassini (TH) aggregation function (Torabi and Hassini, 2008) is used to aggregate three objective functions (W_1, W_2, and W_3). The TH method is highly recommended for maximizing the average overall performance of multi-objective systems because it provides balanced and unbalanced solutions under compromise of the objectives. In addition, the TH aggregation function comprises two components: the minimum satisfaction degree of all objectives (ω_o) in the first term, and the satisfaction degree of each separate objective function ($\mu_i(v)$) in the second term. By adding the satisfaction degree of each objective function into the minimum overall satisfaction degree of all objectives, this method achieves the maximum value of the expected sustainable performance of the system. The first objective function OF_1 of the MOMRPFP model based on the TH method is presented as follows:

$$\begin{aligned}
&\text{Max } OF_1 = \xi\omega_o + (1 - \xi) \sum_{i=1,2,3} h_i \mu_i(v) \\
&\text{S. t} \quad \omega_o \leqslant \mu_i(v),\ i = 1, 2, 3
\end{aligned} \tag{15}$$

where $h_i$and ξ indicate the weight of the ith objective function ($\sum_i h_i$=1) and the coefficient of compensation, respectively. Two values, $\mu_i(v)$ and ω_o, are defined by functions as follows.

$$\begin{aligned}
&\omega_o = \min_{w_i}\{\mu_i(v)\} \\
&\mu_i(v) = \begin{cases} 1 & W_i < W_i^{PIS} \\ \frac{E(W_i^{NIS}) - E(W_i)}{E(W_i^{NIS}) - E(W_i^{PIS})} & W_i^{PIS} \leqslant W_i \leqslant W_i^{NIS},\ \ i = 1, 2, 3 \\ 0 & W_i > W_i^{NIS} \end{cases}
\end{aligned} \tag{16}$$

where $E(W_i)$ indicates the expected function of each objective and $E(W_i^{PIS})$ is the expected positive ideal solution of the ith objective that can be determined by solving the ith expected objective function with system constraints. Its expected negative ideal solution $E(W_i^{NIS})$ can then be estimated as follows:

$$\begin{aligned}
&E(W_1^{NIS}) = E[W_1(x_d^m, q_{id}, q_{is})_2^{PIS}] \\
&E(W_2^{NIS}) = E[W_2(x_d^m, q_{id}, q_{is})_1^{PIS}] \\
&E(W_3^{NIS}) = E[W_3(x_d^m, q_{id}, q_{is})_1^{PIS}]
\end{aligned} \tag{17}$$

To minimize optimality robustness-to minimize the maximum deviation over the expected value of objective functions-the second objective function OF_2 is developed. OF_2 adjusts the degree of satisfaction of each objective function. It is formulated by the maximum value of each objective function minus its minimum satisfaction degree.

$$\begin{aligned}
&\text{Min } OF_2 = \xi\Big(\sum_{i=1,2,3} h_i \mu'_i(v) - \omega_o\Big) \\
&\mu'_i(v) = W_i^{\max},\ i = 1, 2, 3.
\end{aligned} \tag{18}$$

Consider the first original objective function (Min $W_1 = \widetilde{f}\, x_d^m + \widetilde{c}\,(q_{id} + q_{is})$) with triangular fuzzy numbers *f* and *c* with three prominent points (i.e., $\widetilde{f} = (f^{(1)}, f^{(2)}, f^{(3)})$), illustrated in Fig. 4. The maximum value of the first objective function is defined by $\mu'_1(v) = W_1^{max} = f^{(3)} x_d^m + c^{(3)}(q_{id} + q_{is})$. It is similar for the second and the third objective functions.

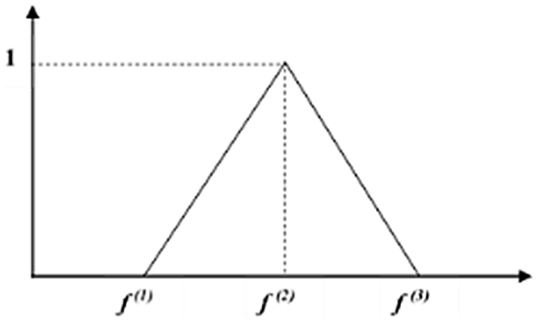


Fig. 4. Triangular possibility distribution of fuzzy number f.

To maximize the feasibility robustness of the proposed model, the third objective function OF_3 is developed as follows:

$$\begin{aligned}\text{Max } OF_3 = \omega[r^{(2)} - \delta(\tfrac{r^{(2)}+r^{(3)}}{2}) + (1-\delta)(\tfrac{r^{(1)}+r^{(2)}}{2})] + \theta[\rho(\tfrac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}) + (1-\rho)(\tfrac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}) - \varphi_d^{m(1)}]x_d^m \\ + \chi[(\tfrac{z^{(1)}+z^{(2)}+z^{(3)}}{3})(1-\alpha)] + \eta[(\tfrac{v^{(1)}+v^{(2)}+v^{(3)}}{3})(1-\beta)]\end{aligned} \tag{19}$$

OF_3 controls the feasibility robustness of the solution vector. The first and second terms are embedded in the model to determine the gap between the worst case and selected values of the uncertain parameters. The other terms determine the violation of each flexible constraint. The parameters ω, χ, η, and θ are representative of the penalty cost for each unit of violation of constraints resulting from imprecision of uncertain parameters. For example, the penalty of traffic congestion in hinterland transport and outsourcing and admitting freight into storage spaces as a result of capacity restrictions are examples of penalty costs incurred at dry ports. The application of penalties facilitates optimizing the value of the minimum satisfaction level of flexible constraints, which eliminates the need for unreliable and direct judgments of decision makers about the value of satisfaction levels. The MOMRPFP model for SDPN optimization can be formulated as follows:

$$\text{Max } OF_1 = \xi\omega_o + (1-\xi)\sum_{w=1,2,3} h_w\mu_w(\upsilon)$$

$$\text{Min } OF_2 = \xi(\sum_{i=1,2,3} h_{w_i}\mu'_{w_i}(\upsilon) - \omega_o)$$

$$\begin{aligned}\text{Max } OF_3 = \omega[r^{(2)} - \delta(\tfrac{r^{(2)}+r^{(3)}}{2}) - (1-\delta)(\tfrac{r^{(1)}+r^{(2)}}{2})] + \theta[\rho(\tfrac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}) + (1-\rho)(\tfrac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}) - \varphi_d^{m(1)}]x_d^m \\ + \chi[(\tfrac{z^{(1)}+z^{(2)}+z^{(3)}}{3})(1-\alpha)] + \eta[(\tfrac{v^{(1)}+v^{(2)}+v^{(3)}}{3})(1-\beta)]x_d^m\end{aligned}$$

$$\text{S. t}$$

$$t_d = \frac{T_d}{\rho(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}) + (1-\rho)(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}) + (\frac{v^{(1)}+v^{(2)}+v^{(3)}}{3})(1-\beta)},$$

$$Aq_{id} + Bq_{is} = [\delta(\tfrac{r^{(2)}+r^{(3)}}{2}) + (1-\delta)(\tfrac{r^{(1)}+r^{(2)}}{2})] - (\tfrac{z^{(1)}+z^{(2)}+z^{(3)}}{3})(1-\alpha),$$

$$Cq_{id} = 0,$$

$$Sq_{id} \leqslant [\rho(\tfrac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}) + (1-\rho)(\tfrac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2})]x_d^m + [(\tfrac{v^{(1)}+v^{(2)}+v^{(3)}}{3})(1-\beta)]x_d^m,$$

$$\omega_o \leqslant \mu_w(\upsilon)\ w = 1, 2, 3,$$

$$Px_d^m \leqslant 1,$$

$$x_d^m \in \{0, 1\},$$

$$q_{id}, q_{is} \geqslant 0,$$

$$\omega_o \geqslant 0, \omega_o \leqslant 1. \tag{20}$$

4.3. The interactive fuzzy solution approach

To solve the MOMRPFP model, we have proposed a fuzzy solution method based on ε-constraint method. The ε-constraint is a well-known approach to solving multi-objective programming models. This approach is able to provide a Pareto optimal curve for decision makers based on the a posteriori or generation multi-objective method (Hwang and Masud, 1979; Mavrotas, 2009). One main advantage of the ε-constraint method is that all solutions of the problem are discovered with a payoff table that is able to provide comprehensive available information to support the selection of the final solution of the decision maker. However, the value selected for ε by a conventional payoff table may lead to dominated or weakly efficient solutions in the Pareto optimal curve; therefore, improved ε-constraint methods have been developed during the past decade. In this study, to implement the modified ε-

constraint method, the satisfaction degree of economic, environmental, and social objectives (see Eq. (15)) is kept in the model objective function, while the adjusted satisfaction degree of the three objective functions of economy, social, and environment (see Eq.18) and the penalty cost function to control the violation of each flexible constraint of capacity and demand (see Eq. (19)) are used as side constraints in the constraints system of the model. The MOMRPFP model can be converted into a single-objective problem based on the modified ε-constraint method as follows.

$$
\begin{gathered}
\text{Max} = OF_1 \\
\text{S. t} \\
OF_2 \leqslant \varepsilon_2, \\
OF_3 \geqslant \varepsilon_3, \\
t_d = \frac{T_d}{\rho(\frac{\varphi_d^{m(1)} + \varphi_d^{m(2)}}{2}) + (1-\rho)(\frac{\varphi_d^{m(2)} + \varphi_d^{m(3)}}{2}) + (\frac{v^{(1)} + v^{(2)} + v^{(3)}}{3})(1-\beta)}, \\
Aq_{id} + Bq_{is} = [\delta(\frac{r^{(2)} + r^{(3)}}{2}) + (1-\delta)(\frac{r^{(1)} + r^{(2)}}{2})] - (\frac{z^{(1)} + z^{(2)} + z^{(3)}}{3})(1-\alpha), \\
Cq_{id} = 0, \\
Sq_{id} \leqslant [\rho(\frac{\varphi_d^{m(1)} + \varphi_d^{m(2)}}{2}) + (1-\rho)(\frac{\varphi_d^{m(2)} + \varphi_d^{m(3)}}{2})]x_d^m + [(\frac{v^{(1)} + v^{(2)} + v^{(3)}}{3})(1-\beta)]x_d^m, \\
\omega_o \leqslant \mu_w(\upsilon)\ w = 1, 2, 3, \\
Px_d^m \leqslant 1, \\
x_d^m \in \{0, 1\},\ q_{id},\ q_{is} \geqslant 0,\ \omega_o \geqslant 0,\ \omega_o \leqslant 1.
\end{gathered}
\tag{21}
$$

Here, ε_2 and ε_3 are the satisfaction levels of the objective functions OF_2 and OF_3, respectively, and the solutions can be achieved by adjusting their parametrical variations in the right sides of the constrained objective functions. The third objective function (OF_3) should be maximized and its related constraints should be in the form of $f_3(x) \geq \varepsilon_3$.

In the ε-constraint solution method, the objective functions OF_2 and OF_3 become the optimality and feasibility constraints limiting the value of the decision variable x_d^m through its duality and to minimize the duality gap, and therefore the solution is optimal. The steps of the proposed solution method for the MOMRPFP model can be summarized in the form of an algorithm, as follows:

Algorithm:

Step 1: Solve the single-objective problem under the MPFRP model by using model (13) to determine the positive ideal solution for each objective function.
Step 2: Determine the negative ideal solution for each objective function by using Eq. (17).
Step 3: Elicit linear membership functions that express the satisfaction degree of the objective functions. Eq. (16) is used in this step.
Step 4: Transform the model into the MOMRPFP model according to the explanation in Section 4.2 and model (20), and identify all relevant parameters provided by decision makers.
Step 5: Build the payoff table as illustrated in Table 1 by using the results of the ith objective function to calculate values for other objective functions.
Step 6: Solve the MOMRPFP model by using the ε-constraint solution method (see model 21) with the epsilon ε_i given by decision makers on the basis of the preference in the payoff table.
Step 7: If the decision maker is satisfied with the current solution, stop. Otherwise, provide another compromise solution by changing the values of ξ and ε_i (and if necessary the value of h_i) and proceed to step 4.

5. Computational experiments and evaluation

According to the relevant literatures (Bairamzadeh et al., 2015; Chang et al., 2015; Feng et al., 2013; Gu and Lam, 2013;

Table 1
Structure of payoff table in ε-constraint method.

Single Objective Problem i	$f_2(x_i)$	.....	$f_p(x_i)$
x_2^*	$f_2(x_2^*)$	.....	$f_p(x_2^*)$
.	.	.....	.
x_p^*	$f_2(x_p^*)$	.....	$f_p(x_p^*)$
W_i^{min}	$minf_2(x_i)$	.....	$minf_p(xi)$
W_i^{max}	$maxf_2(x_i)$	.....	$maxf_p(x_i)$
$\text{minf}_2(x_i) \leq \varepsilon_2 \leq \text{maxf}_2(x_i)$			
.....			
$\text{minf}_p(x_i) \leq \varepsilon_p \leq \text{maxf}_p(x_i)$			

Table 2
Parameters related to CO_2 emission.

Parameters	Fuzzy value
CO_2 emissions for opening and operating of dry port (o^{op})	(0.037, 0.054, 0.078)
CO_2 emissions per distance per container by road (o^{rd})	(0.034, 0.044, 0.054)
CO_2 emissions per distance per container e by rail (o^{rl})	(0.006, 0.007, 0.008)
Unit cost for handling CO_2 emissions (e^{cb})	(0.03, 0.04, 0.05)

Ghodratnama et al., 2015; Pishvaee and Razmi, 2012; Pishvaee et al., 2012), a test instance-based approach is used in this section to validate the performance of the MOMRPFP model. In this study, the SDPN has a single seaport, nine potential locations and three capacity levels for dry ports, and one type of exported freight from thirty-three shippers. The scenario of the problem is the same with the case study of Dalian's port in China in Chang et al. (2015). We have also used the exported freight demand data of shippers from their study.

First, we briefly explain the data used in the example and then present significant results of the test problem based on the efficiency in terms of the computational time required to solve the model and provide efficient solutions for SDPN design in presence of uncertainties. All tests were coded using LINGO 10.0 (LINDO Systems, Chicago, IL, USA) optimization software running on a dual-core 3.40 GHz computer with 8.0 GB of RAM.

5.1. Setting data

In our model, the three objective functions of economic costs, CO_2 emissions handling costs, and social costs are considered. Thus, many qualitative parameters related to environmental and social issues must be identified through reference to related papers and the opinions of experts. Three prominent points regarding triangular fuzzy numbers for the uncertain parameters in the model are also estimated by a group of experts and decision makers. To evaluate environmental impacts of the SDPN, we use unit CO_2 emissions per distance travelled by road and rail transport, which is estimated according to vehicle speed and fuel consumption. We assume that diesel fuel is used for both transport modes and that vehicle speed is affected by traffic congestion. Table 2 shows unit CO_2 emissions and the unit cost for handling CO_2 emissions based on the studies of Lättilä et al. (2013) and Xu et al. (2013). The transportation distance between nodes in the network is identified based on a map of China using Global Positioning System.

For the social terms in our model, the social cost for each unemployed person and immigrant can be obtained from websites or reports related to government social policies. In this paper, we use data from a website in Taiwan *(https://eng.stat.gov.tw)* to define these parameters. Additionally, estimates of the cost of human resources, holding cost for unit cargo, and average congestion time provided by Xu et al. (2013) are used in this study (Table 3). In addition, the parameters used to indicate an area's attractiveness to immigrants and traffic congestion level are defined by decision makers or experts. Finally, the number of jobs created depends on the number of dry ports opened and their capacities.

The other data in the proposed model, such as export demand in shippers, installation costs of candidate dry ports, and capacity levels, are obtained from Chang et al. (2015). The deterministic parameters in this paper are modified into triangular fuzzy numbers according to the opinion of experts. Generally, all remaining uncertainty parameters in our model are treated as triangular fuzzy numbers according to the assumptions and opinions of decision makers.

We set the weight factor h_i for the three economic, environmental, and social objectives to be 0.4, 0.3 and 0.3, respectively. The penalty cost for each unit of violation of constraints regarding cargo export demand and the capacity of dry ports are $\omega = \chi = \eta = \theta = 3.14$. The weighting factors of the cost components in each economic, environmental, and social objective function are $\psi_{eco} = 0.8$ and $\psi_{env} = \psi_{soc} = 0.7$, respectively. All remaining flexible parameters of the proposed model, such as the coefficient of compensation (ξ), the confidence levels of constraints delimiting uncertain parameters (δ and ρ), and the satisfaction levels of flexible constraints (α and β) are shown in Table 4. The impact of varying flexible parameters from 0 to 1 is investigated in Section 5.3.

5.2. SDPN design-related decisions

According to the optimal number, location, and capacity of dry ports illustrated in Table 5, the optimal volume of containers transferred through dry ports is shown in Table 6. As it can be seen from Table 5, there are eight dry ports constructed at the

Table 3
Parameters related to traffic congestion.

Parameters	Fuzzy value
Unit time cost for human resources per time (c^{hr})	(7, 8, 9)
Average congestion time within 70 km range (t^{dl})	(0.4, 0.5, 0.6)
Holding cost for unit cargo (c^{cdm})	(0.05, 0.08, 0.12)

Table 4
Flexible parameters of the MOMRPFP model.

Parameters	Value
Coefficient of compensation (ξ)	0.7
Confidence levels of uncertain constraint about demand (δ)	0.4
Confidence levels of uncertain constraint about capacity (ρ)	0.7
Satisfaction levels of flexible constraints (α and β)	0.7

Table 5
Optimal location and capacity of dry ports.

Location	1	2	3	4	5	6	7	8	9
Capacity (containers)	122,000	142,000	142,000	152,000	122,000	–	152,000	122,000	112,000

Table 6
Optimal number of containers transported in SDPN.

Shippers	x_1^3	x_2^3	x_3^3	x_4^3	x_5^3	x_7^3	x_8^3	x_9^3	Seaport
1	8661								
2						6784			1025
3	1432		2679					2449	
4		7589							
5					22,211				
6			94,866		64,939				
7		13,965		49,281		5153			
8						20,783			
9								65,291	
10					3700			5280	
11		17,332							
12						18,002			
13	10,639								
14				76,562					
15								19,419	
16	5614								988
17		4909		6700					
18								6006	
19							7094		
20		4043							856
21					7496				
22	72,156								
23							16,654	3414	
24					23,654				
25	6802								175
26		5463	34,581						
27				19,457					
28		6227							
29						101,262			
30								3126	
31	16,696		6879				98,252	7015	
32		82,472							
33			2976						
Total	122,000	142,000	141,981	152,000	122,000	151,984	122,000	112,000	3044

x_d^m: d: dry port location; m: capacity of dry port.

maximum capacity level. The optimal number of containers transferred through dry ports accounts for more than 80% of total exported freight of shippers and there are six dry port locations (1, 2, 4, 5, 8, and 9) that reach capacity constraint, as shown in Table 6. Under the optimal solution, the total cost of SDPN is 9.6829E + 08.

5.3. Robustness analysis

In this section, we conduct a sensitivity analysis on both the optimality and feasibility robustness for the test problem to show the

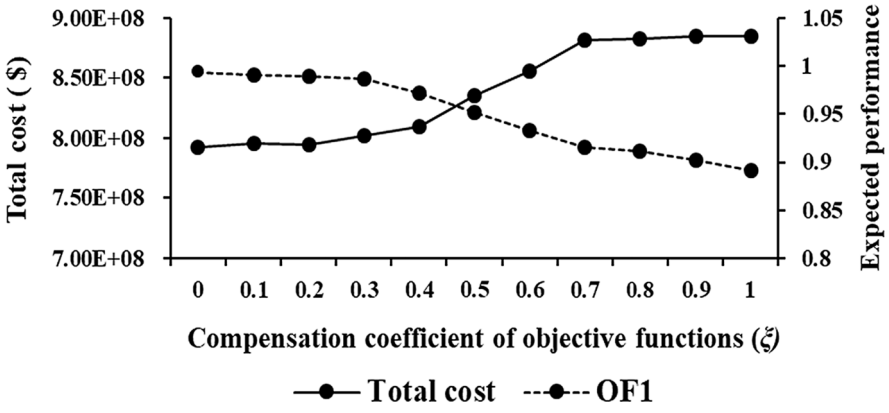


Fig. 5. Total cost and satisfaction degree for different value of ξ.

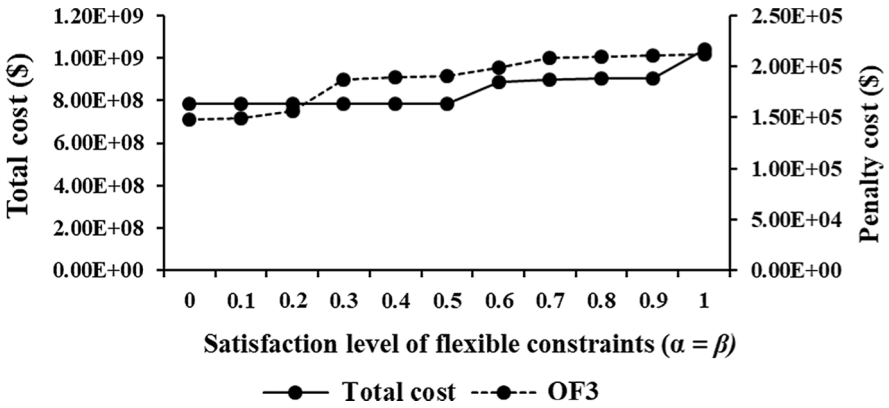


Fig. 6. Total cost and penalty cost for different values of α and β.

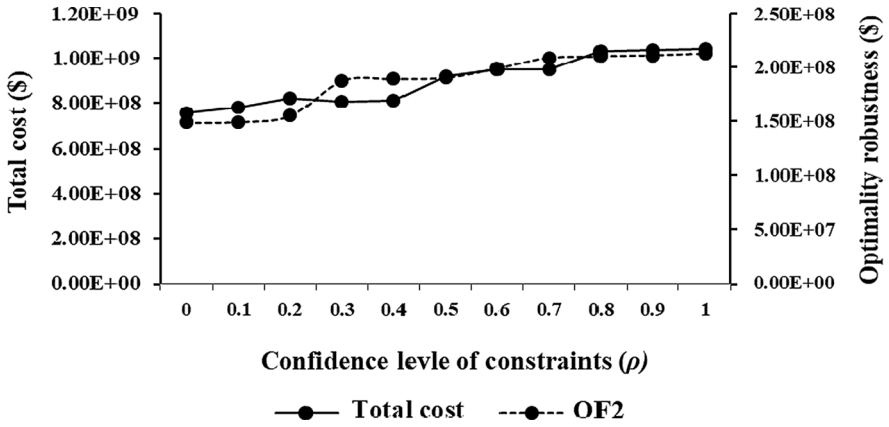


Fig. 7. Total cost and penalty cost for different values of ρ.

behaviour of the proposed model. This model is separately solved for different values of the coefficients of optimality and feasibility robustness (i.e., ξ, α, β, δ and ρ) under the epsilon value range as $OF_2 \leq 1.0E+09$ and $OF_3 \geq 2.0E+05$, referred from the payoff table of the test problem in Appendix E. Fig. 5. illustrates the results in a curve for total costs, including economic, environmental, and social costs (W_1, W_2 and W_3) and the expected performance of the network (OF_1). It is seen that the increase in the value of ξ in the objective function causes an increase in total costs and a decrease in the satisfaction degree of the network. Fig. 6 shows that, with an increase of α and β, the total costs and penalty costs of the network increase. When decision makers respond to flexible constraints with a high satisfaction level, many resources are used to meet demand (i.e., a greater number of dry ports are opened) and capacity

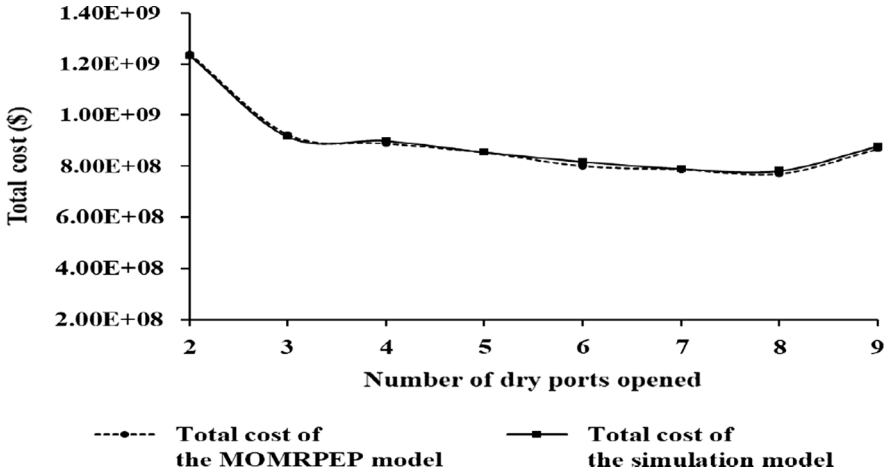


Fig. 8. Total cost comparison of MOMRPFP model with simulation model for different number of dry ports.

levels are also added at each dry port. For a given value of α, β and δ, a larger value of ρ results in a larger value for total costs and high penalty costs, as seen in Fig. 7. It is clear that the performance of the shortage of capacity, which is adjusted with the value of coefficient ρ. In reality, by increasing the β and ρ, the optimality robustness is increased.

5.4. Assessing the performance of the proposed MOMRPEP model

5.4.1. Validation of the proposed MOMRPEP model

In this section, we show that the results of the proposed MOMRPEP model for the total cost of the SDPN, including economic, environmental, and social costs. The proposed MOMRPEP model is validated by comparing simulated and calculated results.

As shown in Fig. 8, total costs of the proposed and simulated model are consistent with different number of dry ports opened from 2 to 9 dry ports. Both models achieve minimal total cost of SDPN when 8 dry ports to be opened. The results indicated that the proposed MOMRPEP model can be used to make SDPN design-related decisions to minimize the total cost.

Also, we conduct a fluctuation analysis of the expected performance (OF_1) with the number of iterations for the test instance to show more explicit evidences about the validity of the proposed MOMRPEP model.

As shown in Fig. 9, the number of required iterations is 37,856 times and CPU time is 5 s. The expected performance value of SDPN begins to reach a steady at the 16,022th iteration. The stability of the expected performance value in later 21,834 (37,856 − 16,022) iterations ensures high accuracy of the SDPN design-related decisions. In addition, the number of dry ports opened is almost unchanged through each iteration after 16,022th iteration. It reveals that the proposed model is capable of making the correct decision for the SDPN design problem under uncertainty situations.

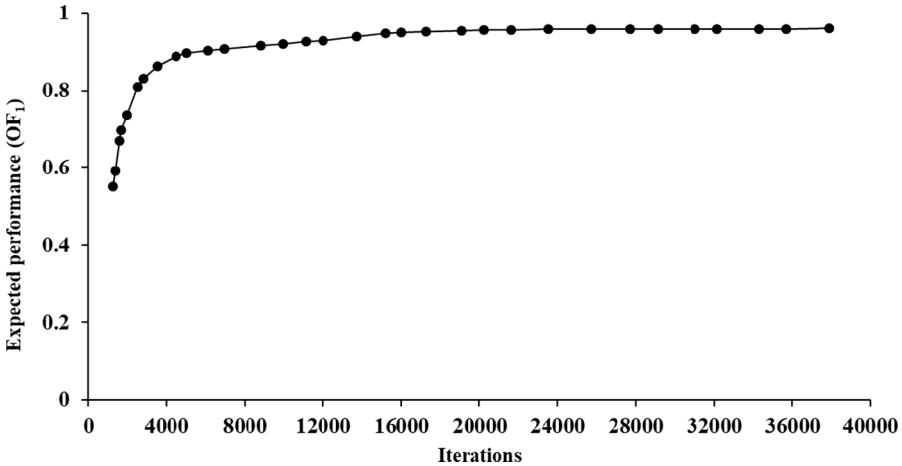


Fig. 9. Fluctuation of the expected performance value with the number of iterations for test instance.

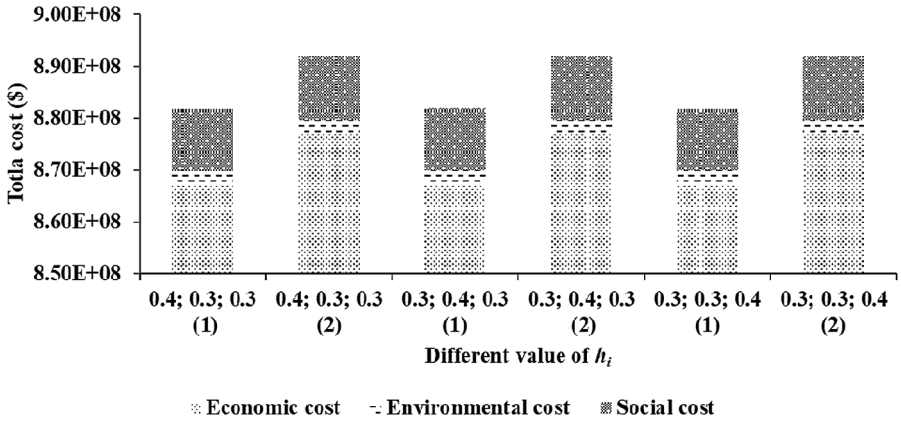


Fig. 10. Total cost comparison of MOMRPFP model with α-parametric model for different weights of objective. 1: MOMRPFP model and 2: α-parametric model.

Table 7
Computational results of the MOMPFRP model and α-parametric model under different weight vectors and epsilon limits.

Model	Weight of objective functions	Epsilon limits	No. of opened dry port	Objective function values			Total cost	CPU time (sec)
				W_1	W_2	W_3		
1	0.4; 0.3; 0.3	$OF_2 \le 1.0E+09$ $OF_3 \ge 2.0E+05$	8	8.6508E+08	2.5424E+06	1.1781E+07	8.7940E+08	6
2	0.4; 0.3; 0.3		8	8.7499E+08	2.6333E+06	1.2233E+07	8.8986E+08	5
1	0.3; 0.4; 0.3	$OF_2 \le 1.0E+09$ $OF_3 \ge 2.0E+05$	8	8.6524E+08	2.5484E+06	1.1957E+07	8.7975E+08	5
2	0.3; 0.4; 0.3		8	8.7499E+08	2.6333E+06	1.2233E+07	8.8986E+08	5
1	0.3; 0.3; 0.4	$OF_2 \le 1.0E+09$ $OF_3 \ge 2.0E+05$	8	8.6722E+08	2.7477E+06	1.1957E+07	8.8192E+08	5
2	0.3; 0.3; 0.4		8	8.7499E+08	2.6333E+06	1.2233E+07	8.8986E+08	4

1: MOMRPFP model and 2: α-parametric model.

Table 8
Average CPU time comparison between MOMRPFP and α-parametric model.

	MOMRPFP model (sec)	α-parametric model (sec)
0.4; 0.3; 0.3	6	5
0.3; 0.4; 0.3	5	5
0.3; 0.3; 0.4	5	4
Average CPU time	**5.33**	**4.67**

Table 9
Size of test problems.

No. of test problem	Shippers (P)	Dry port (D)	Capacity (M)	Seaport (S)	Decision variables	Constraints
Problem 1	100	15	4	1	6160	6396
Problem 2	300	25	5	1	37,925	41,574

5.4.2. Comparison with the α-parametric model

In this section, the MOMRPFP model is compared with the α-parametric model developed by Jimenez et al. (2007). The α-parametric model of SDPNs is detailed in Appendix D. To facilitate the comparison of the two models, we also use the TH aggregation function to aggregate the three objective functions into a single-objective programming of the α-parametric model.

As shown in Fig. 10, the total cost of the SDPN, including economic cost, environmental cost and social cost in the MOMRPFP model is improved about 1,14% compared to the α-parametric model. In Table 7, with the same number of opened dry ports for both models (eight dry ports) but the economic and social costs of the MOMRPFP model are smaller, and the environmental costs are

Table 10
Results of objective functions (OFi) for MOMRPFP model of test problem 1.

Weight of objective functions	Epsilon limits	Overall satisfaction degree	Max OF_1 (Expected performance)	Min OF_2 (Optimality robustness)	Max OF_3 (Feasibility robustness)	No. of opened dry port	W_1 (Economic costs)	W_2 (Environmental impacts)	W_3 (Social costs)	Total cost	CPU time (sec)
0.4; 0.3; 0.3	$OF_2 \le 3.0E+09$ $OF_3 \ge 2.0E+06$	0.94	0.983472	1.3387E+09	2.6966E+06	12	4.8159E+09	3.7234E+07	6.2745E+07	4.9159E+09	80
0.4; 0.3; 0.3	$OF_2 \le 2.5E+09$ $OF_3 \ge 1.5E+06$	0.90	0.951311	1.4269E+09	1.5754E+06	10	5.1197E+09	6.2981E+07	9.2651E+07	5.2753E+09	85
0.4; 0.3; 0.3	$OF_2 \le 2.0E+09$ $OF_3 \ge 1.0E+05$	0.82	0.885298	1.3302E+09	1.3538E+05	7	6.2851E+09	1.0849E+08	1.4013E+08	6.5337E+09	78
0.3; 0.4; 0.3	$OF_2 \le 3.0E+09$ $OF_3 \ge 2.0E+06$	0.96	0.992301	9.9741E+09	2.7889E+06	13	4.7997E+09	2.9835E+07	5.2406E+07	4.8819E+09	82
0.3; 0.4; 0.3	$OF_2 \le 2.5E+09$ $OF_3 \ge 1.5E+06$	0.90	0.953814	1.1364E+09	1.5272E+06	10	5.3872E+09	6.1372E+07	9.0602E+07	5.5392E+09	82
0.3; 0.4; 0.3	$OF_2 \le 2.0E+09$ $OF_3 \ge 1.0E+05$	0.82	0.899949	1.3302E+09	1.3538E+05	7	6.2570E+09	1.0781E+08	1.4013E+08	6.5049E+09	69
0.3; 0.3; 0.4	$OF_2 \le 3.0E+09$ $OF_3 \ge 2.0E+06$	0.99	1.015120	9.2943E+09	2.9425E+06	14	4.4962E+09	1.2955E+07	2.9117E+07	4.5383E+09	70
0.3; 0.3; 0.4	$OF_2 \le 2.5E+09$ $OF_3 \ge 1.5E+06$	0.90	0.941711	1.1398E+09	1.5036E+06	10	5.3974E+09	6.4072E+07	9.3140E+07	5.5546E+09	85
0.3; 0.3; 0.4	$OF_2 \le 2.0E+09$ $OF_3 \ge 1.0E+05$	0.82	0.883115	1.3329E+09	1.3538E+05	7	6.2570E+09	1.0781E+08	1.4013E+08	6.5049E+09	69
Average CPU time									*77.78*		
Aggregate CPU time									*700*		

Table 11
Results of objective functions (OFi) for MOMRPFP model of test problem 2.

Weight of objective functions	Epsilon limits	Overall satisfaction degree	Max OF_1 (Expected performance)	Min OF_2 (Optimality robustness)	Max OF_3 (Feasibility robustness)	No. of opened dry port	W_1 (Economic costs)	W_2 (Environmental impacts)	W_3 (Social costs)	Total cost	CPU time (sec)
0.4; 0.3; 0.3	$OF_2 \le 8.0E+12$ $OF_3 \ge 4.5E+09$	0.98	4.897526	2.7856E+12	4.8572E+09	22	6.9815E+11	4.2347E+09	3.4107E+10	7.3649E+11	245
0.4; 0.3; 0.3	$OF_2 \le 5.0E+12$ $OF_3 \ge 2.5E+09$	0.92	2.978526	3.7580E+12	3.9826E+09	23	7.1719E+11	6.1829E+09	4.1564E+10	7.6494E+11	220
0.4; 0.3; 0.3	$OF_2 \le 6.0E+11$ $OF_3 \ge 8.0E+07$	0.89	2.987525	2.0897E+11	9.5489E+07	18	7.2158E+11	1.0489E+10	8.7586E+10	8.1966E+11	196
0.3; 0.4; 0.3	$OF_2 \le 8.0E+12$ $OF_3 \ge 4.5E+09$	0.98	5.782546	7.4179E+12	4.8898E+09	23	6.7799E+11	3.7425E+09	3.0104E+10	7.1184E+11	251
0.3; 0.4; 0.3	$OF_2 \le 5.0E+12$ $OF_3 \ge 2.5E+10$	0.92	4.968125	3.4631E+12	3.8975E+09	18	7.2387E+11	6.3424E+09	3.9856E+10	7.7007E+11	215
0.3; 0.4; 0.3	$OF_2 \le 6.0E+11$ $OF_3 \ge 8.0E+07$	0.90	3.058925	2.0231E+11	9.3125E+07	15	7.2057E+11	1.1253E+10	8.2056E+10	8.1388E+11	190
0.3; 0.3; 0.4	$OF_2 \le 8.0E+12$ $OF_3 \ge 4.5E+09$	0.98	4.150152	7.3492E+12	4.9475E+09	23	6.4269E+11	4.2347E+09	3.4107E+10	6.8103E+11	250
0.3; 0.3; 0.4	$OF_2 \le 5.0E+12$ $OF_3 \ge 2.5E+10$	0.92	2.945987	3.9813E+12	3.8526E+09	20	7. 4379E+11	6.3625E+09	3.9856E+10	4.6219E+11	228
0.3; 0.3; 0.4	$OF_2 \le 6.0E+11$ $OF_3 \ge 8.0E+07$	0.90	1.989478	2.3239E+11	9.3125E+07	15	7.2057E+11	1.1253E+10	8.7586E+10	8.1941E+11	190
Average											*220.56*
Aggregate CPU time											*1985*

larger. That is explained by the capacity selection for establishing each dry port is dissimilar between two models. In addition, the proposed model achieves similar efficiency about the CPU time index to that of the α-parametric model. That is our approach could solve the problems in a few seconds. As shown in Table 8, the average CPU time of the MOMRPFP model over three runs is 5.33 s, 1.14 times as long as that of the α-parametric model (4.67 s).

5.4.3. Efficiency about the computational time for large-sized instances

To show the performance of the MOMRPFP model for big data size problem, we perform two test problems in Table 9. The big data-size problems considered in our study are the larger than previous DPN design studies.

For both test problems, we set the coefficient of compensation as $\xi = 0.6$, the confidence level of constraints delimiting parameters as $\delta = 0.4$ and $\rho = 0.7$, the satisfaction level of flexible constraints as $\alpha = \beta = 0.7$, the weighting factors as $\psi_{eco} = 0.8$ and $\psi_{env} = \psi_{soc} = 0.7$, and the weight vectors for three objectives, respectively as $h_i = 0.3$, 0.3, and 0.4. In this sections, a sensitivity analysis is conducted for weight vectors and epsilon limits to assess the fluctuation of objective values and CPU time.

The results of the test problem 1, including the overall satisfaction degree, the optimal values of the three objective functions, the number of opened dry ports, and the CPU time, are shown in Table 7. There is a marked fluctuation in the optimal value of the three objectives when the weight vectors change from 0.4, 0.3, and 0.3 to 0.3, 0.3, and 0.4 with the same epsilon limits ($OF_2 \leq 3.0E+09$ and $OF_3 \geq 2.0E+09$). Here, the overall satisfaction degree of the model is highest (0.99%) when the number of opened dry ports is largest (14 dry ports) and the total cost of the network is the smallest. In other words, three objective functions that minimize total costs approaches a decentralized SDPN in which a greater number of dry ports are opened at bigger epsilon limits of constraints for optimality robustness (OF_2) and feasibility robustness (OF_3).

In a multi-objective programming problem related to sustainable development, decisions related to environmental and social objectives clearly have significant effects on the economic objective to obtain a Pareto optimal solution for all objective functions. In this case, the number of opened dry ports is maximized to minimize environmental and social costs; moreover, the economic costs are lowered by reducing the number of containers directly transported to seaport. The proposed approach effectively solves the SDPN design problem in particular and sustainable development-related multi-objective programming problems in general; thus, it has valuable applications in management.

As can be seen from Tables 10 and 11, the average CPU time for nine runs is 77.78 s and 220.56 s for test problems 1 and 2, respectively. Using constraints and decision variables as a proxy for problem size, the aggregate CPU time in test problem 2 is three times higher than that in test problem 1, whereas the number of decision variables and constraints are approximately multiplied by six. The proposed algorithm can solve a large-sized data problem in less than 2 min, as in test problem 1, and less than 4 min in test problem 2. It is obvious that the proposed model can effectively be used not only for designing an DPN in a sustainable and effective manner, but also for any multi-objective programming problem in practice with large data size under an uncertain environment.

6. Conclusions and future research

Although logistic activities are mainly carried out by private operators, governments are still responsible for influencing the environmental and social effects of cargo transportation systems. This research solves the SDPN design problem with consideration of multiple objectives related to economic benefits and environmental and social impact. The strategic and tactical decisions about the optimal number, location, and capacity of dry ports, and the number of containers transferred through dry ports are simultaneously determined to achieve long-term sustainable development goals.

The major contribution of this research is the development of an MOMRPFP approach that extends the mixed robust optimization into possibilistic programming and flexible programming to cope with the multi-objective nature of the real problem and the presence of uncertain parameters in the objective functions and constraints. The proposed MOMRPFP approach is useful for making the optimal decisions with minimum values for the objective functions and efficient in achieving the computational time requirements for large problems. The numerical analyses show the applicability of the proposed solution approach and model for designing complex dry port–based distribution networks with multiple objectives in the uncertain environment of real-life situations.

Regarding the problem considered in this research, future research could address SDPNs with multiple seaports and multiple types of export cargo. In addition, the DPN should be extended for both outbound and inbound distribution. The vehicle routing problem could be integrated into the SDPN design problem to identify reasonable locations for facilities in the network with assessment of the potential harmful effects on the environment and society. The MOMRPFP approach could be applied in many fields involving multiple objectives and uncertainty. Finally, algorithm application-related research, such as that using genetic algorithms and metaheuristic approaches under uncertain environments, should also be conducted in the SDPN design field.

Acknowledgement

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Appendix A

In this appendix we provide detail BMPFP model of the proposed model.

$$\text{Min } W_1 = \psi_{eco}\left(\begin{array}{l}\sum_{d\in D}\sum_{m\in M}(\frac{cf_d^{m(1)}+cf_d^{m(2)}+cf_d^{m(3)}}{3})x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3})f^{id}q_{id} \\ + \sum_{d\in D}\sum_{m\in M}\sum_{i\in P}(\frac{c_d^{str(1)}+c_d^{str(2)}+c_d^{str(3)}}{3})t_d + (\frac{c_d^{handl(1)}+c_d^{handl(2)}+c_d^{handl(3)}}{3})q_{id} \\ + \sum_{d\in D}\sum_{m\in M}(\frac{c^{rl(1)}+c^{rl(2)}+c^{rl(3)}}{3})f^{ds}q_{ds}\end{array}\right)$$

$$+ (1-\psi_{eco})\sum_{i\in P}(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3})f^{is}q_{is}$$

$$\text{Min } W_2 = \psi_{env}\left(\sum_{d\in D}\sum_{m\in M}(\frac{o^{op(1)}+o^{op(2)}+o^{op(3)}}{3})x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3})f^{id}q_{id} + \sum_{d\in D}\sum_{m\in M}(\frac{o^{rl(1)}+o^{rl(2)}+o^{rl(3)}}{3})f^{ds}q_{ds}\right)(\frac{e^{cb(1)}+e^{cb(2)}+e^{cb(3)}}{3})$$

$$+ (1-\psi_{env})\sum_{i\in P}(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3})f^{is}q_{is})(\frac{e^{cb(1)}+e^{cb(2)}+e^{cb(3)}}{3})$$

$$\text{Min } W_3 = \psi_{soc}\left(\begin{array}{l}\sum_{d\in D}\sum_{m\in M}(n^{uep}-(\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3}))s^{uep}x_d^m + \sum_{d\in D}\sum_{m\in M}(\frac{f^{im(1)}+f^{im(2)}+f^{im(3)}}{3})(\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3})s^{im}x_d^m + \\ \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}((\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}) + (\frac{c^{str(1)}+c^{str(2)}+c^{str(3)}}{3})q_{id})(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3})\end{array}\right)$$

$$+ (1-\psi_{soc})\sum_{i\in P}((\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}) + (\frac{c^{str(p)}+c^{str(m)}+c^{str(o)}}{3})q_{is})(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3})$$

S. t

$$t_d = \frac{T_d}{\rho(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}) + (1-\rho)(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}) + (\frac{v^{m(1)}+v^{m(2)}+v^{m(3)}}{3})(1-\beta)};$$

$$\sum_{d\in D}\sum_{m\in M}q_{id} + q_{is} = \delta(\frac{r_i^{(2)}+r_i^{(3)}}{2}) + (1-\delta)(\frac{r_i^{(1)}+r_i^{(2)}}{2}) - (\frac{z^{m(1)}+z^{m(2)}+z^{m(3)}}{3})(1-\alpha),\ \forall\, i\in P$$

$$\sum_{i\in P}q_{id} = q_{ds},\ \forall\, d\in D$$

$$\sum_{i\in P}q_{id} \leqslant [\rho(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}) + (1-\rho)(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}) + (\frac{v^{m(1)}+v^{m(2)}+v^{m(3)}}{3})(1-\beta)]x_d^m,\ \forall\, d\in D, \forall\, m\in M$$

$$\sum_{m\in M}x_d^m \leqslant 1,\ \forall\, d\in D$$

$$x_d^m \in \{0;1\},\ \forall\, d\in D, \forall\, m\in M$$

$$q_{id} \geqslant 0, \forall\, i\in P,\ \forall\, d\in D$$

$$q_{is} \geqslant 0, \forall\, i\in P,\ \forall\, s\in S$$

Appendix B

In this appendix we provide detail MPFRP model of the proposed model.

$$
\begin{aligned}
MinW_1 = \psi_{eco}\Big(& \sum_{d\in D}\sum_{m\in M}\Big(\frac{cf_d^{m(1)}+cf_d^{m(2)}+cf_d^{m(3)}}{3}\Big)x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\Big(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3}\Big)f^{id}q_{id} + \\
& \sum_{d\in D}\sum_{m\in M}\sum_{i\in P}\Big(\frac{c_d^{str(1)}+c_d^{str(2)}+c_d^{str(3)}}{3}\Big)t_d + \Big(\frac{c_d^{handl(1)}+c_d^{handl(2)}+c_d^{handl(3)}}{3}\Big)q_{id} \Big) \\
& + \sum_{d\in D}\sum_{m\in M}\Big(\frac{c^{rl(1)}+c^{rl(2)}+c^{rl(3)}}{3}\Big)f^{ds}q_{ds} \\
& + (1-\psi_{eco})\sum_{i\in P}\Big(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3}\Big)f^{is}q_{is} \\
+ \pi[& (\psi_{eco}\sum_{d\in D}\sum_{d\in D}cf_d^{m(3)}x_d^m + \psi_{eco}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}f^{ds(3)}c^{rd}q_{id} + \psi_{eco}\sum_{d\in D}\sum_{m\in M}\sum_{i\in P}(c_d^{str(3)}t_d + c_d^{handl(3)})q_{id} \\
& + \psi_{eco}\sum_{d\in D}\sum_{m\in M}f^{ds(3)}c^{rl}q_{ds} + (1-\psi_{eco})\sum_{i\in P}f^{ds(3)}c^{rd}q_{is}) \\
& - (\psi_{eco}\sum_{d\in D}\sum_{m\in M}\Big(\frac{cf_d^{m(1)}+cf_d^{m(2)}+cf_d^{m(3)}}{3}\Big)x_d^m + \psi_{eco}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\Big(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3}\Big)f^{id}q_{id} \\
& + \psi_{eco}\sum_{d\in D}\sum_{m\in M}\sum_{i\in P}\Big(\frac{c_d^{str(1)}+c_d^{str(2)}+c_d^{str(3)}}{3}\Big)t_dq_{id} + \psi_{eco}\sum_{d\in D}\sum_{m\in M}\Big(\frac{c^{rl(1)}+c^{rl(2)}+c^{rl(3)}}{3}\Big)f^{ds}q_{ds} \\
& + (1-\psi_{eco})\sum_{i\in P}\Big(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3}\Big)f^{is}q_{is})] \\
+ \omega[& \sum_{i\in P}r_i^{(2)} - \delta\Big(\frac{r_i^{(2)}+r_i^{(3)}}{2}\Big) - (1-\delta)\Big(\frac{r_i^{(1)}+r_i^{(2)}}{2}\Big)] + \theta[\sum_{d\in D}\sum_{m\in M}\rho\Big(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}\Big) + (1-\rho)\Big(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}\Big) - \varphi_d^{m(1)}]x_d^m \\
& + \chi[\sum_{d\in D}\Big(\frac{z^{m(1)}+z^{m(2)}+z^{m(3)}}{3}\Big)(1-\alpha)] + \eta[\sum_{m\in M}\Big(\frac{v^{m(1)}+v^{m(2)}+v^{m(3)}}{3}\Big)(1-\beta)]x_d^m \\
MinW_2 = \psi_{env}(& \sum_{d\in D}\sum_{m\in M}\Big(\frac{o^{op(1)}+o^{op(2)}+o^{op(3)}}{3}\Big)x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\Big(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3}\Big)f^{id}q_{id} \\
& + \sum_{d\in D}\sum_{m\in M}\Big(\frac{o^{rl(1)}+o^{rl(2)}+o^{rl(3)}}{3}\Big)f^{ds}q_{ds}\Big)\Big(\frac{e^{cb(1)}+e^{cb(2)}+e^{cb(3)}}{3}\Big) \\
& + (1-\psi_{env})\sum_{i\in P}\Big(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3}\Big)f^{is}q_{is}\Big)\Big(\frac{e^{cb(1)}+e^{cb(2)}+e^{cb(3)}}{3}\Big) \\
+ \pi[& (\psi_{env}\sum_{d\in D}\sum_{m\in M}o^{op(3)}x_d^m + \psi_{env}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}o^{rd(3)}f^{id}q_{id} + \psi_{env}\sum_{d\in D}\sum_{m\in M}f^{ds}o^{rl(3)}q_{ds} \\
& + (1-\psi_{env})\sum_{i\in P}f^{ds}o^{rd(3)}q_{is})e^{cb(3)} \\
& - (\psi_{env}\sum_{d\in D}\sum_{m\in M}\Big(\frac{o^{op(1)}+o^{op(2)}+o^{op(3)}}{3}\Big)x_d^m + \psi_{env}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\Big(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3}\Big)f^{id}q_{id} \\
& + \psi_{env}\sum_{d\in D}\sum_{m\in M}\Big(\frac{o^{rl(1)}+o^{rl(2)}+o^{rl(3)}}{3}\Big)f^{ds}q_{ds} \\
& + (1-\psi_{env})\sum_{i\in P}\Big(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3}\Big)f^{is}q_{is}\Big)\Big(\frac{e^{cb(1)}+e^{cb(2)}+e^{cb(3)}}{3}\Big))] \\
+ \omega[& \sum_{i\in P}r_i^{(2)} - \delta\Big(\frac{r_i^{(2)}+r_i^{(3)}}{2}\Big) - (1-\delta)\Big(\frac{r_i^{(1)}+r_i^{(2)}}{2}\Big)] + \theta[\sum_{d\in D}\sum_{m\in M}\rho\Big(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}\Big) + (1-\rho)\Big(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}\Big) - \varphi_d^{m(1)}]x_d^m \\
& + \chi[\sum_{d\in D}\Big(\frac{z^{m(1)}+z^{m(2)}+z^{m(3)}}{3}\Big)(1-\alpha)] + \eta[\sum_{m\in M}\Big(\frac{v^{m(1)}+v^{m(2)}+v^{m(3)}}{3}\Big)(1-\beta)]x_d^m \\
MinW_3 = \psi_{soc}\sum_{d\in D}\sum_{m\in M}\Big(n^{uep} - \Big(\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3}\Big)\Big)s^{uep}x_d^m + \psi_{soc}\sum_{d\in D}\sum_{m\in M}\Big(\frac{f^{im(1)}+f^{im(2)}+f^{im(3)}}{3}\Big)\Big(\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3}\Big)s^{im}x_d^m + \\
\psi_{soc}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\Big(\Big(\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}\Big) + \Big(\frac{c^{str(1)}+c^{str(2)}+c^{str(3)}}{3}\Big)q_{id}\Big)\Big(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3}\Big) \\
+ (1-\psi_{soc})\sum_{i\in P}\Big(\Big(\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}\Big) + \Big(\frac{c^{str(1)}+c^{str(2)}+c^{str(3)}}{3}\Big)q_{is}\Big)\Big(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3}\Big) \\
+ \pi[& (\psi_{soc}\sum_{d\in D}\sum_{m\in M}(n^{uep} - n^{ep(3)})s^{uep}x_d^m + \psi_{soc}\sum_{d\in D}\sum_{m\in M}f^{im(3)}n^{ep(3)}s^{im}x_d^m + \psi_{soc}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}(c^{hr(3)} + c^{str(3)})t^{dl(3)}q_{id} \\
& + (1-\psi_{soc})\sum_{i\in P}(c^{hr(3)} + t^{dl(3)})t^{dl(3)}q_{is}) \\
& - (\psi_{soc}\sum_{d\in D}\sum_{m\in M}\Big(n^{uep} - \Big(\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3}\Big)\Big)s^{uep}x_d^m + \psi_{soc}\sum_{d\in D}\sum_{m\in M}\Big(\frac{f^{im(1)}+f^{im(2)}+f^{im(3)}}{3}\Big)\Big(\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3}\Big)s^{im}x_d^m + \\
& \psi_{soc}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\Big(\Big(\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}\Big) + \Big(\frac{c^{str(1)}+c^{str(2)}+c^{str(3)}}{3}\Big)q_{id}\Big)\Big(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3}\Big) \\
& + (1-\psi_{soc})\sum_{i\in P}\Big(\Big(\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}\Big) + \Big(\frac{c^{str(1)}+c^{str(2)}+c^{str(3)}}{3}\Big)q_{is}\Big)\Big(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3}\Big)] \\
+ \omega[& \sum_{i\in P}r_i^{(2)} - \delta\Big(\frac{r_i^{(2)}+r_i^{(3)}}{2}\Big) - (1-\delta)\Big(\frac{r_i^{(1)}+r_i^{(2)}}{2}\Big)] + \theta[\sum_{d\in D}\sum_{m\in M}\rho\Big(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}\Big) + (1-\rho)\Big(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}\Big) - \varphi_d^{m(1)}]x_d^m \\
& + \chi[\sum_{d\in D}\Big(\frac{z^{m(1)}+z^{m(2)}+z^{m(3)}}{3}\Big)(1-\alpha)] + \eta[\sum_{m\in M}\Big(\frac{v^{m(1)}+v^{m(2)}+v^{m(3)}}{3}\Big)(1-\beta)]x_d^m
\end{aligned}
$$

$$\text{S. t}$$

$$t_d = \frac{T_d}{\rho(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2})+(1-\rho)(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2})+(\frac{v^{m(1)}+v^{m(2)}+v^{m(3)}}{3})(1-\beta)};$$

$$\sum_{d\in D}\sum_{m\in M} q_{id} + q_{is} = \delta(\frac{r_i^{(2)}+r_i^{(3)}}{2})+(1-\delta)(\frac{r_i^{(1)}+r_i^{(2)}}{2})-(\frac{z^{m(1)}+z^{m(2)}+z^{m(3)}}{3})(1-\alpha),\ \forall\, i\in P$$

$$\sum_{i\in P} q_{id} = q_{ds},\ \forall\, d\in D$$

$$\sum_{i\in P} q_{id} \leqslant [\rho(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2})+(1-\rho)(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2})+(\frac{v^{m(1)}+v^{m(2)}+v^{m(3)}}{3})(1-\beta)]x_d^m,\ \forall\, d\in D, \forall\, m\in M$$

$$\sum_{m\in M} x_d^m \leqslant 1,\ \forall\, d\in D$$

$$x_d^m \in \{0;1\},\ \forall\, d\in D, \forall\, m\in M$$

$$q_{id} \geqslant 0, \forall\, i\in P,\ \forall\, d\in D$$

$$q_{is} \geqslant 0, \forall\, i\in P,\ \forall\, s\in S$$

Appendix C

In this appendix we provide detail MOMRPFP model for sustainable seaport and dry port network.

$$\text{Max } OF_1 = \xi\omega_o + (1-\xi)(h_{w_1}((E(W_1^{NIS}) - (\psi_{eco}\sum_{d\in D}\sum_{m\in M}(\frac{cf_d^{m(1)}+cf_d^{m(2)}+cf_d^{m(3)}}{3})x_d^m +$$

$$\psi_{eco}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3})f^{id}q_{id} + \psi_{eco}\sum_{d\in D}\sum_{m\in M}\sum_{i\in P}(\frac{c_d^{str(1)}+c_d^{str(2)}+c_d^{str(3)}}{3})t_d + (\frac{c_d^{handl(1)}+c_d^{handl(2)}+c_d^{handl(3)}}{3})q_{id} +$$

$$\psi_{eco}\sum_{d\in D}\sum_{m\in M}(\frac{c^{rl(1)}+c^{rl(2)}+c^{rl(3)}}{3})f^{ds}q_{ds} + (1-\psi_{eco})\sum_{i\in P}(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3})f^{is}q_{is}))/(E(W_1^{NIS}) - E(W_1^{PIS}))) +$$

$$h_{w2}((E(W_2^{NIS}) - ((\psi_{env}\sum_{d\in D}\sum_{m\in M}(\frac{o^{op(1)}+o^{op(2)}+o^{op(3)}}{3})x_d^m + \psi_{env}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3})f^{id}q_{id}$$

$$+ \psi_{env}\sum_{d\in D}\sum_{m\in M}(\frac{o^{rl(1)}+o^{rl(2)}+o^{rl(3)}}{3})f^{ds}q_{ds}$$

$$+ (1 - \psi_{env})\sum_{i\in P}(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3})f^{is}q_{is})(\frac{e^{cb(1)}+e^{cb(2)}+e^{cb(3)}}{3})))/(E(W_2^{NIS}) - E(W_2^{PIS}))) +$$

$$h_{w3}((E(W_3^{NIS}) - (\psi_{soc}\sum_{d\in D}\sum_{m\in M}(n^{uep} - (\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3}))s^{uep}x_d^m + \psi_{soc}\sum_{d\in D}\sum_{m\in M}(\frac{f^{im(1)}+f^{im(2)}+f^{im(3)}}{3})(\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3})s^{im}x_d^m +$$

$$\psi_{soc}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}((\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}) + (\frac{c^{str(1)}+c^{str(2)}+c^{str(3)}}{3})q_{id})(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3})$$

$$+ (1 - \psi_{soc})\sum_{i\in P}((\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}) + (\frac{c^{str(1)}+c^{str(2)}+c^{str(3)}}{3})q_{is})(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3})))/(E(W_3^{NIS}) - E(W_3^{PIS})))$$

$$\text{Min } OF_2 = \xi(h_{w_1}(\psi_{eco}\sum_{d\in D}\sum_{d\in D} cf_d^{m(3)}x_d^m + \psi_{eco}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M} f^{id}c^{rd(3)}q_{id} + \psi_{eco}\sum_{d\in D}\sum_{m\in M}\sum_{i\in P}(c_d^{str(3)}t_d + c_d^{handl(3)})q_{id}$$

$$+ \psi_{eco}\sum_{d\in D}\sum_{m\in M} f^{ds}c^{rl(3)}q_{ds} + (1-\psi_{eco})\sum_{i\in P} f^{is}c^{rd(3)}q_{is}) +$$

$$h_{w2}((\psi_{env}\sum_{d\in D}\sum_{m\in M} o^{op(3)}x_d^m + \psi_{env}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M} f^{ds(3)}o^{rd}q_{id} + \psi_{env}\sum_{d\in D}\sum_{m\in M} f^{ds(3)}o^{rl}q_{ds} + (1-\psi_{env})\sum_{i\in P} f^{ds(3)}o^{rd}q_{is})e^{cb(3)}) +$$

$$h_{w3}(\psi_{soc}\sum_{d\in D}\sum_{m\in M}(n^{uep} - n^{ep(3)})s^{uep}x_d^m + \psi_{soc}\sum_{d\in D}\sum_{m\in M} f^{im(3)}n^{ep(3)}s^{im}x_d^m$$

$$+ \psi_{soc}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}(c^{hr(3)} + c^{str(3)})t^{dl(3)}q_{id} + (1-\psi_{soc})\sum_{i\in P}(c^{hr(3)} + t^{dl(3)})t^{dl(3)}q_{is})) - \omega_o)$$

$$\text{Max } OF_3 = \omega[\sum_{i\in P} r_i^{(2)} - \delta(\frac{r_i^{(2)}+r_i^{(3)}}{2}) - (1-\delta)(\frac{r_i^{(1)}+r_i^{(2)}}{2})] + \theta[\sum_{d\in D}\sum_{m\in M}\rho(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}) + (1-\rho)(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}) - \varphi_d^{m(1)}]x_d^m$$

$$+ \chi[\sum_{d\in D}(\frac{z^{m(1)}+z^{m(2)}+z^{m(3)}}{3})(1-\alpha)] + \eta[\sum_{m\in M}(\frac{v^{m(1)}+v^{m(2)}+v^{m(3)}}{3})(1-\beta)]x_d^m$$

S. t

$$\omega_o \leqslant (E(W_1^{NIS}) - (\psi_{eco} \sum_{d\in D}\sum_{m\in M}(\frac{cf_d^{m(1)} + cf_d^{m(2)} + cf_d^{m(3)}}{3})x_d^m + \psi_{eco}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}(\frac{c^{rd(1)} + c^{rd(2)} + c^{rd(3)}}{3})f^{id}q_{id}$$
$$+ \psi_{eco}\sum_{d\in D}\sum_{m\in M}\sum_{i\in P}(\frac{c_d^{str(1)} + c_d^{str(2)} + c_d^{str(3)}}{3})t_d + (\frac{c_d^{handl(1)} + c_d^{handl(2)} + c_d^{handl(3)}}{3})q_{id} + \psi_{eco}\sum_{d\in D}\sum_{m\in M}(\frac{c^{rl(1)} + c^{rl(2)} + c^{rl(3)}}{3})f^{ds}q_{ds}$$
$$+ (1-\psi_{eco})\sum_{i\in P}(\frac{c^{rd(1)} + c^{rd(2)} + c^{rd(3)}}{3})f^{is}q_{is}))/(E(W_1^{NIS}) - E(W_1^{PIS})),$$

$$\omega_o \leqslant (E(W_1^{NIS}) - ((\sum_{d\in D}\sum_{m\in M}(\frac{o^{op(1)} + o^{op(2)} + o^{op(3)}}{3})x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}(\frac{o^{rd(1)} + o^{rd(2)} + o^{rd(3)}}{3})f^{id}q_{id}$$
$$+ \psi_{env}\sum_{d\in D}\sum_{m\in M}(\frac{o^{rl(1)} + o^{rl(2)} + o^{rl(3)}}{3})f^{ds}q_{ds}$$
$$+ (1-\psi_{env})\sum_{i\in P}(\frac{o^{rd(1)} + o^{rd(2)} + o^{rd(3)}}{3})f^{is}q_{is})(\frac{e^{cb(1)} + e^{cb(2)} + e^{cb(3)}}{3})))/(E(W_1^{NIS}) - E(W_1^{PIS})),$$

$$\omega_o \leqslant (E(W_1^{NIS}) - (\psi_{soc}\sum_{d\in D}\sum_{m\in M}(n^{uep} - (\frac{n^{ep(1)} + n^{ep(2)} + n^{ep(3)}}{3}))s^{uep}x_d^m + \psi_{soc}\sum_{d\in D}\sum_{m\in M}(\frac{f^{im(1)} + f^{im(2)} + f^{im(3)}}{3})(\frac{n^{ep(1)} + n^{ep(2)} + n^{ep(3)}}{3})s^{im}x_d^m +$$
$$\psi_{soc}\sum_{i\in P}\sum_{d\in D}\sum_{m\in M}((\frac{c^{hr(1)} + c^{hr(2)} + c^{hr(3)}}{3}) + (\frac{c^{str(1)} + c^{str(2)} + c^{str(3)}}{3})q_{id})(\frac{t^{dl(1)} + t^{dl(2)} + t^{dl(3)}}{3})$$
$$+ (1-\psi_{soc})\sum_{i\in P}((\frac{c^{hr(1)} + c^{hr(2)} + c^{hr(3)}}{3}) + (\frac{c^{str(1)} + c^{str(2)} + c^{str(3)}}{3})q_{is})(\frac{t^{dl(1)} + t^{dl(2)} + t^{dl(3)}}{3})))/(E(W_1^{NIS}) - E(W_1^{PIS})),$$

$$t_d = \frac{T_d}{\rho(\frac{\varphi_d^{m(1)} + \varphi_d^{m(2)}}{2}) + (1-\rho)(\frac{\varphi_d^{m(2)} + \varphi_d^{m(3)}}{2}) + (\frac{v^{(1)} + v^{(2)} + v^{(3)}}{3})(1-\beta)},$$

$$Aq_{id} + Bq_{is} = [\delta(\frac{r^{(2)} + r^{(3)}}{2}) + (1-\delta)(\frac{r^{(1)} + r^{(2)}}{2})] - (\frac{z^{(1)} + z^{(2)} + z^{(3)}}{3})(1-\alpha),$$

$$Cq_{id} = 0,$$

$$Sq_{id} \leqslant [\rho(\frac{\varphi_d^{m(1)} + \varphi_d^{m(2)}}{2}) + (1-\rho)(\frac{\varphi_d^{m(2)} + \varphi_d^{m(3)}}{2})]x_d^m + [(\frac{v^{(1)} + v^{(2)} + v^{(3)}}{3})(1-\beta)]x_d^m,$$

$$\omega_o \leqslant \mu_w(\upsilon)\ w = 1, 2, 3,$$

$$Px_d^m \leqslant 1,$$

$$x_d^m \in \{0, 1\},$$

$$q_{id},\ q_{is} \geqslant 0,$$

$$\omega_o \geqslant 0,\ \omega_o \leqslant 1.$$

Appendix D

In this appendix we provide detail α-parametric model for sustainable seaport and dry port network.

$$\text{Min } W_1 = \psi_{eco}\left(\sum_{d\in D}\sum_{m\in M}\left(\frac{cf_d^{m(1)}+cf_d^{m(2)}+cf_d^{m(3)}}{3}\right)x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\left(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3}\right)f^{id}q_{id} + \sum_{d\in D}\sum_{m\in M}\sum_{i\in P}\left(\frac{c_d^{str(1)}+c_d^{str(2)}+c_d^{str(3)}}{3}\right)t_d + \left(\frac{c_d^{handl(1)}+c_d^{handl(2)}+c_d^{handl(3)}}{3}\right)q_{id} + \sum_{d\in D}\sum_{m\in M}\left(\frac{c^{rl(1)}+c^{rl(2)}+c^{rl(3)}}{3}\right)f^{ds}q_{ds}\right) + (1-\psi_{eco})\sum_{i\in P}\left(\frac{c^{rd(1)}+c^{rd(2)}+c^{rd(3)}}{3}\right)f^{is}q_{is}$$

$$\text{Min } W_2 = \psi_{env}\left(\sum_{d\in D}\sum_{m\in M}\left(\frac{o^{op(1)}+o^{op(2)}+o^{op(3)}}{3}\right)x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\left(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3}\right)f^{id}q_{id} + \sum_{d\in D}\sum_{m\in M}\left(\frac{o^{rl(1)}+o^{rl(2)}+o^{rl(3)}}{3}\right)f^{ds}q_{ds}\right)\left(\frac{e^{cb(1)}+e^{cb(2)}+e^{cb(3)}}{3}\right) + (1-\psi_{env})\sum_{i\in P}\left(\frac{o^{rd(1)}+o^{rd(2)}+o^{rd(3)}}{3}\right)f^{is}q_{is}\left(\frac{e^{cb(1)}+e^{cb(2)}+e^{cb(3)}}{3}\right)$$

$$\text{Min } W_3 = \psi_{soc}\left(\sum_{d\in D}\sum_{m\in M}\left(n^{uep}-\left(\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3}\right)\right)s^{uep}x_d^m + \sum_{d\in D}\sum_{m\in M}\left(\frac{j^{im(1)}+j^{im(2)}+j^{im(3)}}{3}\right)\left(\frac{n^{ep(1)}+n^{ep(2)}+n^{ep(3)}}{3}\right)s^{im}x_d^m + \sum_{i\in P}\sum_{d\in D}\sum_{m\in M}\left(\left(\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}\right)+\left(\frac{c^{str(1)}+c^{str(2)}+c^{str(3)}}{3}\right)q_{id}\right)\left(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3}\right)\right) + (1-\psi_{soc})\sum_{i\in P}\left(\left(\frac{c^{hr(1)}+c^{hr(2)}+c^{hr(3)}}{3}\right)+\left(\frac{c^{str(p)}+c^{str(m)}+c^{str(o)}}{3}\right)q_{is}\right)\left(\frac{t^{dl(1)}+t^{dl(2)}+t^{dl(3)}}{3}\right)$$

S. t

$$t_d = \frac{T_d}{\alpha\left(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}\right)+(1-\alpha)\left(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}\right)};$$

$$\sum_{d\in D}\sum_{m\in M} q_{id} + q_{is} = \alpha\left(\frac{r_i^{(1)}+r^{(2)}}{2}\right)+(1-\alpha)\left(\frac{r^{(2)}+r^{(3)}}{2}\right),\ \forall\, i\in P$$

$$\sum_{i\in P} q_{id} = q_{ds},\ \forall\, d\in D$$

$$\sum_{i\in P} q_{id} \leqslant \left[\alpha\left(\frac{\varphi_d^{m(1)}+\varphi_d^{m(2)}}{2}\right)+(1-\alpha)\left(\frac{\varphi_d^{m(2)}+\varphi_d^{m(3)}}{2}\right)\right]x_d^m,\ \forall\, d\in D, \forall\, m\in M$$

$$\sum_{m\in M} x_d^m \leqslant 1,\ \forall\, d\in D$$

$$x_d^m \in \{0;1\},\ \forall\, d\in D, \forall\, m\in M$$

$$q_{id}\geqslant 0, \forall\, i\in P,\ \forall\, d\in D$$

$$q_{is}\geqslant 0, \forall\, i\in P,\ \forall\, s\in S$$

Appendix E

In this appendix we provide results of positive ideal solution (PIS) and negative ideal solution (NIS) for objective functions (W_1, W_2 and W_3) that solved under the single objective mixed possibilistic-flexible robust programming model. In addition, we also provide the payoff table to identify epsilon limit and objective functions (OF_2 and OF_3) for solving multi-objective mixed possibilistic-flexible robust programming model (see Tables 12–17).

Table 12
Results of PIS and NIS for objective functions (W_i) of test problem 1.

Objective function	Positive ideal solution (PIS)	Negative ideal solution (NIS)	CPU time (sec)
E(W_1)	8.12415E + 08	3.75351E + 09	03
E(W_2)	1.71548E + 06	9.48899E + 07	02
E(W_3)	8.18692E + 06	1.17960E + 09	01

Table 13
Payoff table for objective functions under ε-constraint method of test problem 1.

Single objective problem i	OF_2	OF_3
$MinOF_2(x_d^m, q_{id}, q_{is})$	2.11944E + 08	2.16657E + 05
$MaxOF_3(x_d^m, q_{id}, q_{is})$	1.02886E + 09	3.10830E + 04
2.11944E + 08 ≤ ε_2 ≤ 1.02886E + 09 3.10830E + 04 ≤ ε_3 ≤ 2.16657E + 05		

Table 14
Results of PIS and NIS for objective functions (Wi) of test problem 2.

Objective function	Positive ideal solution (PIS)	Negative ideal solution (NIS)	CPU time (sec)
$E(W_1)$	4.32308E + 09	1.12356E + 10	54
$E(W_2)$	7.18547E + 06	2.86395E + 08	15
$E(W_3)$	2.10097E + 07	3.36322E + 08	17

Table 15
Payoff table for objective functions under ε-constraint method of test problem 2.

Single objective problem i	OF_2	OF_3
$MinOF_2(x_d^m, q_{id}, q_{is})$	1.1813E + 09	2.0159E + 06
$MaxOF_3(x_d^m, q_{id}, q_{is})$	3.0914E + 09	9.4246E + 04
1.1813E + 09 ≤ ε_2 ≤ 3.0914E + 09 9.4246E + 04 ≤ ε_3 ≤ 2.0159E + 06		

Table 16
Results of PIS and NIS for objective functions (Wi) of test problem 3.

Objective function	Positive ideal solution (PIS)	Negative ideal solution (NIS)	CPU time (sec)
$E(W_1)$	7.50894E + 11	3.25647E + 12	89
$E(W_2)$	5.25785E + 09	8.97856E + 10	75
$E(W_3)$	3.54786E + 10	7.85642E + 10	75

Table 17
Payoff table for objective functions under ε-constraint method of test problem 3.

Single objective problem i	OF_2	OF_3
$MinOF_2(x_d^m, q_{id}, q_{is})$	5.4512E + 11	4.6125E + 09
$MaxOF_3(x_d^m, q_{id}, q_{is})$	8.4526E + 12	7.5264E + 07
5.4512E + 11 ≤ ε_2 ≤ 8.4526E + 12 7.5264E + 07 ≤ ε_3 ≤ 4.6125E + 09		

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